Expectations versus Reality in Business Formation*

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Abstract

Using administrative data on 17 million U.S. business applications linked to outcomes, we compare potential entrants' expectations about employer entry and first-year employment with realizations. On average, applicants overestimate employment, mainly because many expect to enter but do not. Among those who expect and achieve entry, employment is typically underestimated. Expected employment predicts entry and realized employment, but conditional on entry, realized employment rises less than one-for-one with expectations. Expectation errors are highly heterogeneous and systematically related to application characteristics and local economic conditions, and they predict near-term employment outcomes. A model with informative pre-application signals, heterogeneous priors, learning, and pre-entry selection rationalizes these patterns. Relative to a common-prior counterfactual, the uninformative component of beliefs generates net *excess* expected entry (8.5–9.5% of applications) but net *forgone* actual entry (0.5–0.9% of applications), with the latter roughly doubling over 2017–2021.

Keywords: business applications, expectation errors, uncertainty, entry, employment, misallocation

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1 Introduction

Expectations about the future are central to models of firm behavior and entrepreneurial decision-making. Entrepreneurs form beliefs about whether they will launch a business, how many employees they will have, and how the venture will evolve, shaping initial commitments, resource acquisition, financing, and organizational planning. Because these decisions affect the allocation of inputs, expectation errors can distort entry and resource allocation: optimism may induce excessive entry and misallocation, while pessimism may deter productive ventures. Yet, despite their central role in theory and empirical relevance, there is relatively little systematic evidence on expectation errors at early stages of business formation.

Most studies of firms' expectations focus on businesses already in operation (Guiso and Parigi (1999); Coibion, Gorodnichenko, and Kumar (2018); Bloom, Codreanu, and Fletcher (2025)). Early-stage ventures, however, face greater constraints, frictions, and uncertainty, making expectation errors potentially systematic and large at entry (Camerer and Lovallo (1999); Koellinger, Minniti, and Schade (2007)). These errors may involve both over- and underestimation, with distinct patterns at the extensive margin of entry and the intensive margin of initial employment. Expectations at formation may also reveal unobserved pre-entry heterogeneity relevant for subsequent firm growth (Sterk, Sedláček, and Pugsley (2021)).

Classic models of firm dynamics typically assume that firms form beliefs rationally given available information (e.g., Jovanovic (1982); Hopenhayn (1992)).[1] In Jovanovic's model, firms enter uncertain about their fixed productivity and learn about it over time; in Hopenhayn's framework, productivity is unknown ex ante, observed upon entry, and then evolves stochastically. Behavioral models instead allow expectation errors arising from optimism, bounded rationality, miscalibration, or overconfidence,[2] while other work emphasizes financing frictions and regulatory barriers.[3] Although a rich literature studies entrepreneurial beliefs using surveys and experiments,[4] few studies provide observational evidence linking ex-ante expectations to ex-post outcomes at the business formation stage, partly because such expectations are rarely recorded near entry and linked to

[1] Other firm dynamics frameworks with similar assumptions include Asplund and Nocke (2006), Nocke (2006), and Ericson and Pakes (1995).

[2] See, e.g., Camerer and Lovallo (1999); Koellinger et al. (2007).

[3] See, e.g., Evans and Jovanovic (1989); Djankov, La Porta, Lopez-de Silanes, and Shleifer (2002).

[4] See, e.g., Manski (2004); Astebro, Herz, Nanda, and Weber (2014); Puri and Robinson (2007).

subsequent outcomes.

This paper provides novel observational evidence on entrepreneurial expectations and outcomes using administrative microdata on about 17 million U.S. new-business applications from 2017–2021. The data underlie the Census Bureau's Business Formation Statistics (BFS), originate from IRS Form SS-4 applications for Employer Identification Numbers (EINs), and cover nearly all economically meaningful new-business applications, including all employer businesses, corporations, and partnerships. They report expected initial employment and link to the Census Bureau's Longitudinal Business Database (LBD), allowing us to observe employer entry and track realized employment over time.[5] We use this linkage to study employment expectations reported at application.

Our analysis proceeds in several parts. First, we document broad patterns of forecast error. Applicants overestimate initial employment on average, with substantial heterogeneity: many expect entry but do not enter, some enter unexpectedly, and those who both expect and achieve entry tend to underestimate. Errors arise at both the extensive (entry) and intensive (employment) margins. Mincer–Zarnowitz regressions indicate biased and inefficient forecasts: realized employment falls short of expectations overall, and increases less than one-for-one with expected employment. This pattern is robust across specifications addressing non-linearity and selection in entry. Year-by-year estimates also show a weakening relationship between expected and realized employment from 2017 to 2021, especially after 2019 as the COVID-19 pandemic unfolded.

Second, we relate forecast errors to application attributes and local conditions, without a causal interpretation. Corporations and LLCs report more conservative and better-aligned expectations than sole proprietors or partnerships, consistent with greater planning capacity, formality, or access to advisors.[6] Higher tract-level college share and greater local firm density are associated with less overestimation, while higher non-white share and greater dispersion in incumbent employment and productivity are associated with more, consistent with differences in planning resources, information networks, agglomeration, and uncertainty.[7] Sectoral differences are also pronounced: high-tech and manufacturing applicants exhibit greater overestimation, potentially reflecting greater uncertainty,

[5] non-employer businesses are economically important but generate no paid employment. Because our focus is employment generation, we treat non-employer formation and no business formation equivalently. Dinlersoz, Kroff, Luque, and Novik (2025) studies both employer and non-employer entry originating from business applications.

[6] See Levine and Rubinstein (2017); Hurst and Pugsley (2011).

[7] See Gennaioli, La Porta, Lopez-de Silanes, and Shleifer (2013); Moretti (2004b); Carlino and Kerr (2009); Bachmann, Elstner, and Sims (2013); Wu and Knott (2006).

reliance on intangibles, and longer development cycles, whereas more standardized service sectors such as professional services and real estate exhibit smaller errors. Overall, expectation errors are related to market structure, industry characteristics, and the local business environment.

Third, we examine how expectation errors evolved around COVID-19. Errors began to rise in 2019, as the 12-month window used to assess whether applications became employer businesses increasingly overlapped with the pandemic. As the shock approached, expected employment declined only modestly, while realized employment fell much more sharply, leading to increasing overestimation. After March 2020, expectations stabilized, but realized employment remained depressed, so the gap between expectations and outcomes persisted. The dispersion of forecast errors followed a different pattern: it had already been declining before 2019 and narrowed further around the pandemic onset, partly because more applicants expected zero employment. Dispersion then widened during the recovery, especially among applicants with positive employment expectations, as outcomes became more heterogeneous. Overall, the pandemic illustrates how a large, unexpected macroeconomic shock can affect both the average size and the dispersion of expectation errors.

Fourth, we relate expectation errors to a near-term outcome: third-year employment, with non-entrants and exits coded as zero. Controlling for realized first-year employment and other observables, greater initial overestimation predicts higher third-year employment. In models without pre-application information about business quality (e.g., (Jovanovic, 1982; Hopenhayn, 1992)), forecast errors should have no predictive content once initial outcomes are controlled. Their predictive content instead points to private information not fully revealed by first-year employment.

Motivated by these findings, we extend the learning framework of Jovanovic (1982) by allowing potential entrants to observe informative pre-application signals and to hold heterogeneous priors with an uninformative idiosyncratic component. The informative component links forecasts to latent business quality and makes initial forecast errors predictive of later employment. The uninformative component, together with selection into application, generates aggregate overestimation, while selection into entry produces underestimation among entrants when overestimation is not too large. This combination also delivers a Mincer–Zarnowitz slope strictly between zero and one.

We then derive closed-form measures of the net and gross distortions generated by the uninformative component of beliefs at both the expected-entry and actual-entry margins. The two margins have opposite signs. Relative to a common-prior counterfactual, heterogeneous priors generate net

excess expected entry equal to about 8.5–9.5% of applications, but net *forgone* actual employer entry of about 0.5–0.9% of applications. The latter roughly doubles in magnitude between 2017 and 2021 as forecast efficiency declines. The distinction arises because belief noise changes selection into application: it can induce relatively low-quality potential entrants to apply while deterring some higher-quality potential entrants who otherwise would have become employers. Thus, noisy beliefs can increase planned entrepreneurial activity while reducing realized employer formation. This distinction is particularly salient toward the end of the sample, when business applications surged while forecast efficiency declined and the estimated actual-entry distortion increased.

Our contributions are threefold. First, we extend the literature on early-stage business activity by documenting expectation errors near business formation, distinguishing entry from employment margins (Bayard, Dinlersoz, Dunne, Haltiwanger, Miranda, and Stevens (2018); Guzman and Stern (2020); Dinlersoz, Dunne, Haltiwanger, and Penciakova (2023)). We show how errors vary with applicant characteristics, local and industry conditions, and the COVID-19 shock, contributing to firm entry and business dynamism literature (Decker, Haltiwanger, Jarmin, and Miranda (2014); Akcigit and Ates (2021); Ma and Yang (2024)). We also quantify entry distortions and show that, as forecast efficiency declines, belief noise can reduce realized employer entry even as applications surge. Second, we provide large-scale observational evidence relevant to classical and behavioral theories of entrepreneurship, complementing experimental and survey-based work. Systematic overestimation at formation may help explain engagement in entrepreneurship despite low risk-adjusted returns (Moskowitz and Vissing-Jørgensen (2002); Astebro et al. (2014)). Third, we show that application-stage expectations reveal ex-ante heterogeneity and predict ex-post performance, helping assess near-term job-creation potential before outcomes are realized. Overall, the analysis highlights the importance of early-stage expectations, which remain underexplored in both theory and empirics.

2 Conceptual framework

We study potential entrants, some of whom file a business application in period t to form a new business (employer or non-employer).[8] In the data, applicants reveal expectations about whether

[8]Employer businesses are those that hire workers, as opposed to non-employer businesses, which are operated without paid employees.

they will start an employer business, which we refer to as "entry", and the highest employment their businesses will achieve within a finite horizon of τ periods (e.g., months or quarters). Actual entry and size that are realized over this horizon may diverge from initial expectations. Some applications may become non-employer businesses—an important source of income for many individuals. Since non-employer businesses generate no employment, we do not distinguish them from no formation and classify both as no entry/zero employment based on our definition of entry.

2.1 Entry and employment outcomes

Let V_{it} denote a latent net-value index associated with forming a business during the horizon from t to $t+\tau$. Employer entry occurs when the index exceeds the entry threshold. Denoting the threshold as c_E, the realized entry indicator is

$$e_{it} = \mathbf{1}\{V_{it} > c_E\}. \tag{1}$$

Values of V_{it} at or below c_E correspond to no business formation or to a non-employer business, both of which are coded as no entry.

Let m_{it} denote maximum observed employment during the horizon. By construction,

$$m_{it} \begin{cases} > 0, & \text{if } e_{it} = 1, \\ = 0, & \text{if } e_{it} = 0. \end{cases} \tag{2}$$

Thus, zero employment represents the absence of employer entry during the observation window, while positive employment combines entry with the scale of employment conditional on entry.

2.2 Expectations at the time of application

At the time of application, the potential entrant has an information set $\mathcal{I}_{it}$ and forms a point forecast of latent employment over the horizon,

$$\tilde{\mu}^*_{it} = \tilde{\mathbb{E}}_i \left[m^*_{it} \mid \mathcal{I}_{it} \right], \tag{3}$$

where $\tilde{\mathbb{E}}_i$ denotes the expectation under the applicant's subjective beliefs and m^*_{it} is latent employment, a continuous measure defined whether or not employer entry occurs. Section 5 provides a Gaussian model for this object.

We assume that an applicant applies when her subjective expectation of the net value of the

business exceeds c_A, $\tilde{\mathbb{E}}_i[V_{it} \mid \mathcal{I}_{it}] > c_A$, and plans employer entry when it exceeds c_E, $\tilde{\mathbb{E}}_i[V_{it} \mid \mathcal{I}_{it}] > c_E$. The threshold c_A reflects the cost of applying and exploring the business idea, and c_E the additional cost of launching operations with paid employees, so that $c_A < c_E$. We further assume that V_{it} is measured in employment units, in the sense that $\tilde{\mathbb{E}}_i[V_{it} - m^*_{it} \mid \mathcal{I}_{it}] = 0$.[9] Then $\tilde{\mathbb{E}}_i[V_{it} \mid \mathcal{I}_{it}] = \tilde{\mu}^*_{it}$, and both decisions can be written as thresholds on the employment forecast:

$$\tilde{a}_{it} = \mathbf{1}\{\tilde{\mu}^*_{it} > c_A\}, \qquad \tilde{e}_{it} = \mathbf{1}\{\tilde{\mu}^*_{it} > c_E\},$$

where $\tilde{a}_{it}$ and $\tilde{e}_{it}$ indicate application and planned employer entry. Since c_E is also the threshold for realized entry in (1), an applicant plans employer entry exactly when she expects the net value of the business to clear the entry threshold.

The reported expected employment is

$$\tilde{\mu}_{it} = \tilde{e}_{it}\tilde{\mu}^*_{it}. \tag{4}$$

Thus, an applicant reports zero expected employment when employer entry is not planned and reports the anticipated employment scale when entry is planned. With $c_E \geq 0$, $\tilde{e}_{it} = \mathbf{1}\{\tilde{\mu}_{it} > 0\}$.

The reported expected employment, $\tilde{\mu}_{it}$, need not equal the subjective expectation of zero-inclusive realized employment, $\tilde{\mathbb{E}}_i[m_{it} \mid \mathcal{I}_{it}]$, which weights the employment scale by the subjective probability of entry. This distinction is relevant because Form SS-4 records a planned employment level rather than a subjective probability distribution over entry and employment outcomes.

2.3 Expectation errors

Define the realized entry expectation (or forecast) error as

$$\omega_{it} = \tilde{e}_{it} - e_{it}. \tag{5}$$

Here, $\omega_{it} = 1$ if the potential entrant expected to enter but did not, $\omega_{it} = -1$ if entry occurred despite no expectation of entry, and $\omega_{it} = 0$ if expectations and outcomes match. Similarly, define the realized expectation (or forecast) error of employment as

$$\delta_{it} = \tilde{\mu}_{it} - m_{it}. \tag{6}$$

In this case, $\delta_{it} > 0$ indicates overestimation, $\delta_{it} < 0$ underestimation, and $\delta_{it} = 0$ perfect accuracy.

[9] This is a normalization of the location and scale of V_{it} when the value of the business is affine in latent employment.

2.4 Decomposition of expectation errors in the population

Denote the population shares of expectation-outcome pairs by $s(\tilde{e}, e)$, for $\tilde{e}, e \in \{0, 1\}$. Let $\bar{\delta}(\tilde{e}, e)$ be the population mean of employment expectation error conditional on group $(\tilde{e}, e)$. Then, the overall population mean employment expectation error can be decomposed as

$$\bar{\delta} = \bar{\delta}(1,1)s(1,1) + \bar{\delta}(1,0)s(1,0) + \bar{\delta}(0,1)s(0,1) + \bar{\delta}(0,0)s(0,0), \tag{7}$$

where $\bar{\delta}(0,0) = 0$ by construction. Similarly, the population mean of entry expectation error is $\bar{\omega} = s(1,0) - s(0,1)$. The decomposition (7) is useful in understanding the behavior of expectation errors in different sub-groups of the population.

The empirical analysis uses the sample analogues, $\hat{\bar{\delta}}$ and $\hat{\bar{\omega}}$, computed from observed expected and realized employment and entry. With the large administrative sample, these sample moments converge to their population counterparts under the Strong Law of Large Numbers, given standard regularity conditions.[10] The sample statistics can be used to test for systematic bias in expectations.

3 Data

Our analysis uses U.S. Census Bureau Business Formation Statistics (BFS) microdata, which link business applications to subsequent outcomes. We summarize key features below; see Bayard et al. (2018) for details.

3.1 Business applications

The BFS compiles statistics on early-stage entrepreneurial activity based on applications for Employer Identification Numbers (EINs) via IRS Form SS-4.[11] The underlying microdata contain rich application-level information, including six-digit NAICS industry codes and geographic identifiers. These fields enable identification of *business applications* – EIN applications with a business intent.

The data encompass all economically significant business applications. All employer businesses are required to obtain an EIN to file payroll taxes. Corporations and partnerships, regardless of

[10] The Strong Law requires that sample averages converge almost surely to their expectations. Sufficient conditions include finite first and second moments and, in the non-*i.i.d.* case, restrictions on the nature of the correlations. In our context, expectation errors are unlikely to be strictly *i.i.d.* and may exhibit cross-sectional dependence arising from local or industry shocks or correlated beliefs.

[11] See the BFS website for additional documentation.

employment status, must also acquire one. While sole proprietors without employees do not need an EIN, many obtain one for banking, licensing, or compliance (Fairlie, Kroff, Miranda, and Zolas (2023)). non-employers with EINs represent a substantial share of total non-employer revenues.[12]

Each application includes information on business name/location, date of filing (week), intended start date, reason for applying (e.g., starting a new business, purchasing an existing business, banking purposes), legal form, prior EIN filing, principal business activity, planned date of first wage payment, and the maximum number of employees expected in the next 12 months. These variables capture salient features of the applicant's plans and provide predictive signals about subsequent employer status (Bayard et al. (2018)).[13]

Business application and employer entry are distinct stages, and applicants face substantial uncertainty about whether and when they will hire. Although obtaining an EIN is quick, it is typically needed to open a business bank account, register an LLC or corporation, and obtain business licenses, so applicants often file well before hiring. About half of applicants who plan to hire report a first wage date after the application quarter. Among applications that become employers within four (eight) quarters, the average time to entry is 1.35 (2.21) quarters; these conditional averages understate the delay for applicants as a whole.

The analysis focuses on business applications that report an intent to "start a new business" (by checking the relevant box in the application form for the reason for applying). These constitute 90% of all business applications in the period of analysis. Excluded are those applications with reasons such as banking purposes, purchasing an existing business or changing organizational form, which may not face the same type of uncertainty as new businesses.

The analysis sample contains monthly data over 2017m4–2021m12. While BFS microdata are available since 2004, the expected maximum employment variable is available only from April 2017. The end of the sample period (2021m12) was determined by the availability of actual employment information for applications at the time of our analysis. The analysis sample contains nearly 17 million applications with a plan to start a new business.

[12] See Davis, Haltiwanger, Krizan, Jarmin, Miranda, Nucci, and Sandusky (2009), who also document that EIN non-employer firms report significantly higher revenues on average than their non-EIN counterparts.

[13] The BFS program uses these characteristics to project near-term employer business formations for recent periods not yet observable in the LBD.

3.2 Business formations

The BFS links EIN applications to the Longitudinal Business Database (LBD), a comprehensive panel containing data on firm identifiers, location, industry, age, and quarterly employment and payroll. Using this link, we identify transitions to employer status (entry) by matching application EINs to first observed instances of positive employment in the LBD.

Employment is observed at quarterly frequency. Consistent with the horizon for expected maximum employment, we set the forward-looking horizon for actual employment generation to $\tau = 4$ quarters (12 months) including the quarter of filing. Entry occurs if a firm generates positive employment in any of these quarters. Since application dates vary within the quarter, applicants filing later in a quarter have less time to generate employment within a fixed 4-quarter horizon. To address this timing issue, we use the week of application within a quarter to adjust the window of observation to identify the first positive employment. Specifically, applications filed in week $w = 1, \ldots, 13$ have a probability, $w/13$, of being assigned an additional (fifth) quarter for employment generation. This procedure gives all applicants, on average, a full 12-month window to transition into employer status, mitigating bias from within-quarter timing variation.

3.3 Measures of expected entry, realized entry, and initial size

Form SS–4 asks, "What is the highest number of employees expected in the next 12 months? (Enter 0 if none)," a question recorded for all EIN applications beginning in 2017m4.[14] We construct two expectation measures: (i) expected employer entry, $\tilde{e}_{it}$, equal to one if any employees are expected, and (ii) expected maximum employment, $\tilde{\mu}_{it}$, capturing the intensive margin, with $\tilde{e}_{it} = 1$ when $\tilde{\mu}_{it} > 0$. Form SS–4 also asks for the "first date wages or annuities were paid (month, day, year)." Unlike expected employment, this date is not capped at a 12-month horizon, allowing us to measure expected entry over longer horizons. The expected-entry indicator is nearly identical to the indicator for planned wage payments within the first year (correlation 0.996), suggesting that applicants report consistent expectations about both employment and hiring decisions.

The IRS collects this information primarily for tax administration, such as establishing payroll tax filing requirements (quarterly Form 941 or, for eligible small employers that elect it, annual

[14] To reduce the influence of outliers, we drop observations with $\tilde{\mu}_{it}$ above 100 employees; fewer than one percent of observations exceed this threshold.

Form 944) and related withholding obligations. Applicants attest, under penalties of perjury, that the information is true, correct, and complete to the best of their knowledge and belief, so the reported figure is a good-faith estimate rather than an enforceable commitment. Because these responses have limited immediate financial consequences and can be updated as the business evolves, they provide a useful window into entrepreneurs' initial beliefs. Most applicants complete SS-4 early, often before facing binding regulatory thresholds or external scrutiny from lenders, landlords, or investors, making reported employment plans less likely to reflect strategic considerations than later disclosures. Finally, EIN issuance depends on submitting a complete and valid application rather than performance-based screening, reducing concerns that observed expectations reflect selection through the approval process.

Realized (actual) entry and employment measures are constructed based on the LBD. Entry occurs, i.e., $e_{it} = 1$, if any of the four post-application quarters record positive employment. That is, a horizon of $\tau = 4$ periods (quarters) is used to match the potential entrant's one-year horizon for expectations. Maximum realized employment, m_{it}, is defined as the highest quarterly employment observed during this window. Applications with no observed employment receive a value of zero – indicating no employment generation.[15]

A potential concern is that the LBD measures employment quarterly, while applicants may interpret the 12-month horizon at a finer frequency, such as monthly or bi-weekly. Our realized maximum employment could therefore understate applicants' intended measure even without systematic expectation bias.[16] To address this mismatch, we construct conservative theoretical upper bounds for the expected maximum under monthly ($\tau = 12$) and approximately bi-weekly ($\tau = 24$) measurement, as detailed in Appendix B. Because these bounds exceed the true expected maximum and need not be tight, they provide a conservative test. Our findings remain robust, largely because employment is highly persistent within firms: conditional on entry, the average pairwise correlation across quarterly employment levels is nearly 90%, and likely higher at monthly frequency. This persistence supports using quarterly data to approximate maximum employment over the 12-month horizon.

[15] Similar to the case with the expected maximum employment, we drop observations with realized maximum employment above 100 employees – the remaining cases again representing more than 99% of the observations.

[16] The expected maximum of n i.i.d. random variables is non-decreasing in n. Although this need not hold for non-i.i.d. variables, the expected maximum can still increase with n.

To examine expected and realized entry over horizons other than 12 months, we use the planned first wage payment date reported in each application to measure expected entry. This variable enables us to identify the expected quarter of entry and to compute cumulative expected entry rates by quarter following application. We then compare these with cumulative realized entry rates over time, constructed using the actual quarter of the first wage payment.

A substantial share of applications report neither a planned wage payment date nor positive expected employment; many likely reflect non-employer intent or delayed hiring, yet a small but nontrivial share later become employers. Applications reporting a planned wage payment date—classified as Applications with Planned Wages (WBA) in the BFS—have much higher employer transition rates (27% within four quarters of application, versus 7% for all applications in our sample) and, as noted above, almost always report positive expected employment. Because positive expected employment may indicate more concrete employer plans, expectation errors may differ for this group. Conversely, zero expected employment may yield smaller errors because actual entry and positive employment are less common. We therefore analyze both the full sample, including zero expected employment, and the sample with positive expected employment, which is nearly identical to WBA, and document the differences.

4 Analysis

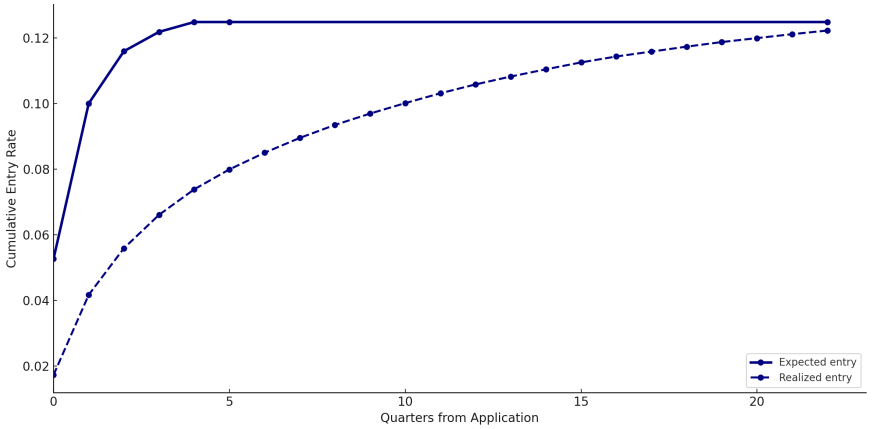
We begin by exploring whether potential entrants have systematic errors in their entry and employment expectations at the time of application.

4.1 Expectations and realizations of entry and initial size

Figure 1 compares cumulative expected and realized entry rates for all applications. Expected entry rises rapidly and plateaus near 12.5% by quarter 6, after which applicants no longer report anticipated first wage payments. Realized entry rises more gradually and continues beyond quarter 6, partially closing the gap. This pattern indicates overprediction of near-term employer formation, despite some delayed entry, and a persistent tendency to overestimate entry likelihood (Puri and Robinson (2007); Koellinger et al. (2007)).

Figure 2 plots average realized against average expected employment 12 months after application

Figure 1: Expected versus realized cumulative entry rate – All applications



Notes: The graph shows expected and realized cumulative entry rate over time starting from the application quarter (corresponding to 0 in the x-axis). All applications between 2017m4-2021m12 are included.

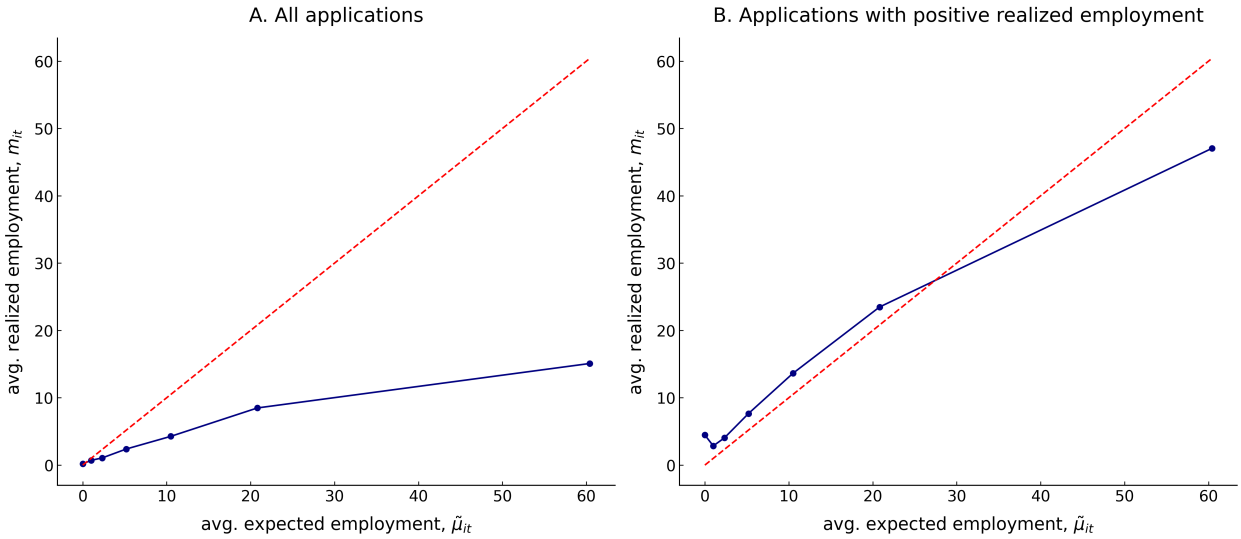
across bins of expected employment. Panel A uses the full sample; Panel B restricts to applications with positive realized employment. The 45-degree line provides a benchmark. In Panel A, realized employment is below expectations throughout the distribution, indicating widespread overestimation, although realized size rises with expectations. In Panel B, realized employment generally exceeds expectations among actual entrants, except in the top bin. Underestimation at the left-most point, where expected employment is zero, arises because only negative errors are possible, though their magnitude is unrestricted. The remaining points show that forecast errors reflect not only entry failures but also underprediction among applicants with lower expected employment. Overall, both the extensive margin of entry and the intensive margin of size contribute to gaps between expectations and outcomes.

4.1.1 Components of the expectation error

We next examine the components of the expectation error based on (7). Table 1 reports average expected employment ($\tilde{\mu}_{it}$), realized employment (m_{it}), and their difference (δ_{it}) across groups defined by expected and actual entry outcomes. It also includes group shares.

Most applications (84.1%) neither expect nor achieve entry. These applications likely represent plans to start non-employer businesses. Only 3.4% of applications both expect entry and enter, while 8.9% expect entry but do not realize it. Another 3.6% enter despite expecting not to. The average entry expectation error is $\hat{\bar{\omega}} = 0.089 - 0.036 = 0.053$, and the total misprediction rate is

Figure 2: Average expected versus realized employment within percentiles of expected employment



Notes: Each point shows the average realized employment against average expected employment within bins of expected employment. The bins correspond to point 0, and the intervals defined by pseudo-percentiles (pct): (0,50th pct), [50th pct,75th pct), [75th pct,90th pct), [90th pct,95th pct), [95th pct,99th pct), and [99th pct,100th pct]. The dashed line is the 45-degree line.

$|0.089| + |-0.036| = 0.125$.

In the full sample, applicants expect to hire 0.53 employees on average but realize only 0.39, yielding a mean overprediction of 0.14. Disaggregating by entry outcomes reveals sharper differences. Those who expect to enter but do not ($\{\tilde{e}, e\} = \{1, 0\}$) overpredict by 4.087 employees. Conversely, those who unexpectedly enter ($\{0, 1\}$) underpredict by 4.462 on average.

Among those who both expect and achieve entry ($\{1, 1\}$), the average error is –1.903, indicating underprediction overall. Yet, this group is quite heterogeneous: 47% underpredict by 6.237 employees, 24% overpredict by 3.872, and 29% forecast employment perfectly–the latter has a small average expected (and realized) employment (about 2 employees) compared to the other two groups, reflecting the fact that the scale of small business operations may be easier to predict. For the broader group (12.3% of the sample) who expect entry ($\{1, \cdot\}$), the average error is 2.431 – resulting from both overestimation by those who fail to enter and those that enter and underestimate.[17]

[17] This group corresponds to the pairs $\{1, 1\}$ and $\{1, 0\}$ in Table 1.

Table 1: Expectation error and its decomposition

	Sample average of:			
Sample	$\tilde{\mu}_{it}$	m_{it}	δ_{it}	% of N
$\{\tilde{e}, e\} = \{\cdot, \cdot\}$	0.530	0.388	0.142	100%
	(0.001)	(0.001)	(0.001)	
$\{\tilde{e}, e\} = \{1, 1\}$	4.779	6.683	-1.903	3.4%
	(0.011)	(0.014)	(0.011)	
$\hat{\bar{\delta}}_t(1,1) < 0$	4.788	11.030	-6.237	1.6%
	(0.014)	(0.026)	(0.018)	
$\hat{\bar{\delta}}_t(1,1) > 0$	8.233	4.361	3.872	0.8%
	(0.032)	(0.021)	(0.019)	
$\hat{\bar{\delta}}_t(1,1) = 0$	1.988	1.988	0.000	1.0%
	(0.007)	(0.007)	(0.000)	
$\{\tilde{e}, e\} = \{1, 0\}$	4.087	0.000	4.087	8.9%
	(0.006)	(0.000)	(0.006)	
$\{\tilde{e}, e\} = \{0, 1\}$	0.000	4.462	-4.462	3.6%
	(0.000)	(0.011)	(0.011)	
$\{\tilde{e}, e\} = \{0, 0\}$	0.000	0.000	0.000	84.1%
	(0.000)	(0.000)	(0.000)	

Notes: Standard errors are in parentheses. N = 16,820,000–rounded for disclosure avoidance. $\delta_{it} = \tilde{\mu}_{it} - m_{it}$.

Overall, expectation errors reflect both extensive-margin gaps (entry failures and surprises) and intensive-margin deviations among entrants, with significant heterogeneity even within the group that correctly forecast entry. The coexistence of over- and underestimation is unlikely to stem solely from bureaucratic conservatism or strategic understatement (e.g., to reduce perceived tax exposure). If such motives were dominant, we would expect aggregate and widespread understatement. The substantial and bidirectional variation is instead consistent with models linking entrepreneurial

outcomes to diverse expectations and strategic orientations.[18]

4.1.2 Decomposition of the variance of expectation errors

Table 2 decomposes the variance of the employment expectation error as

$$Var(\delta_{it}) = Var(\tilde{\mu}_{it}) + Var(m_{it}) - 2Cov(\tilde{\mu}_{it}, m_{it}). \tag{8}$$

In the full sample, error variance is modest (12.69), reflecting the prevalence of zero expected and actual employment. Expected employment accounts for 75% of error variance, realized employment for 68%, and the covariance term for −43%. Expectations and realizations are positively but only modestly correlated (0.30). Among applications with positive expected employment, error variance rises to 69.37; the expectation share increases to 88%, the realization share falls to 58%, the covariance share is −46%, and the correlation rises slightly to 0.32. Overall, error variation is more closely tied to expectations than outcomes, while expectations remain only modestly correlated with realizations even among applicants anticipating employment.

Table 2: Decomposition of the variance of the expectation error

Sample	$Var(\delta_{it})$	$\frac{Var(\tilde{\mu}_{it})}{Var(\delta_{it})}$	$\frac{Var(m_{it})}{Var(\delta_{it})}$	$\frac{-2\,Cov(\tilde{\mu}_{it},m_{it})}{Var(\delta_{it})}$	$Corr(\tilde{\mu}_{it}, m_{it})$
All applications	12.69	0.75	0.68	-0.43	0.30
Positive Exp. ($\tilde{\mu}_{it} > 0$)	69.37	0.88	0.58	-0.46	0.32

Notes: The functions $Var(\cdot)$, $Cov(\cdot,\cdot)$, and $Corr(\cdot,\cdot)$ give the *sample* variance, covariance, and correlation of their arguments, respectively. "Positive Exp." refers to applications that report positive expected maximum employment.

4.2 Expectation errors and application characteristics

Next, we examine heterogeneity in expectation errors across selected application characteristics. These comparisons are unconditional; later regressions control for application characteristics and other observables to estimate partial correlations, without a causal interpretation.

[18] See, e.g., Kihlstrom and Laffont (1979); Lazear (2005); Baron (2004); Astebro, Jeffrey, and Adomdza (2007); Parker (2009); Hurst and Pugsley (2011); Manso (2016).

4.2.1 Legal form of organization

Table 3 summarizes expected and realized employment and expectation errors by legal form of organization (LFO), for all applications and those with positive expected employment. We group all LLCs separately and classify non-LLCs as sole proprietorships, partnerships, or corporations. Expectation errors vary substantially across organizational forms.

Among all applications, corporations and LLCs have the highest average expected and realized employment, consistent with greater formality and growth orientation. Sole proprietorships have the lowest realized employment and relatively large average errors, while corporations have much smaller errors. Among applications with positive expected employment, average errors exceed two employees for every legal form. Thus, even for more sophisticated forms such as corporations and LLCs, plans remain substantially misaligned with outcomes.

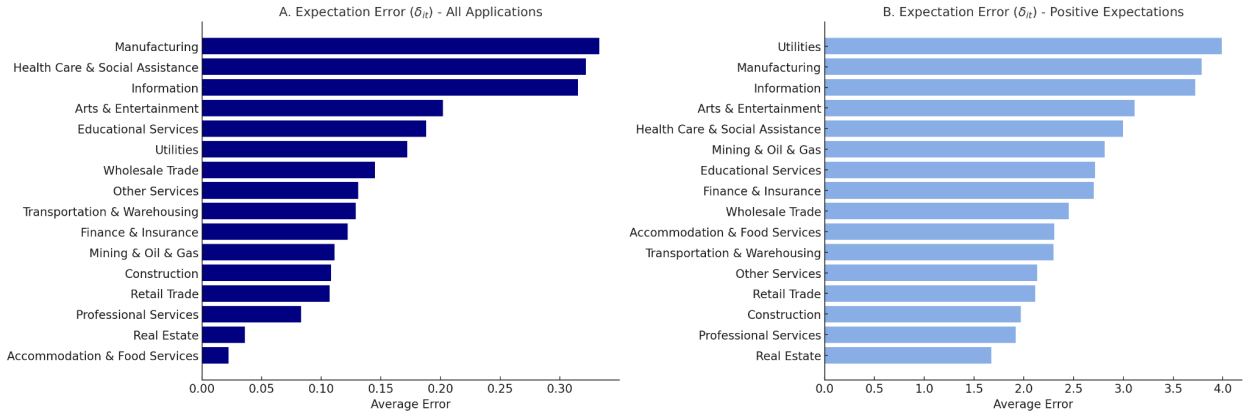
Table 3: Summary statistics for expectation errors by legal form of organization

	All applications			Positive expectations ($\tilde{\mu}_{it} > 0$)		
Legal form	$\tilde{\mu}_{it}$	m_{it}	δ_{it}	$\tilde{\mu}_{it}$	m_{it}	δ_{it}
Sole proprietorship	0.220	0.058	0.161	3.522	1.317	2.205
	(0.001)	(0.000)	(0.001)	(0.003)	(0.001)	(0.003)
Partnership	0.453	0.231	0.222	4.511	1.901	2.610
	(0.001)	(0.001)	(0.001)	(0.007)	(0.003)	(0.008)
Corporation	0.916	0.842	0.075	6.217	3.673	2.544
	(0.002)	(0.003)	(0.002)	(0.007)	(0.005)	(0.008)
LLC	0.521	0.377	0.144	4.763	2.218	2.545
	(0.001)	(0.001)	(0.001)	(0.004)	(0.002)	(0.005)
Other	0.508	0.197	0.311	5.246	2.065	3.181
	(0.002)	(0.001)	(0.002)	(0.013)	(0.005)	(0.014)

Notes: Entries are sample means by legal form of organization. LLCs are classified under one category and non-LLCs are presented in 4 distinct groups (sole proprietorship, corporations, partnerships, others). "Positive expectations" refers to applications reporting positive expected maximum employment. Standard errors are in parentheses. $\delta_{it} = \tilde{\mu}_{it} - m_{it}$.

Differences across organizational forms may reflect variation in information access, venture complexity, planning capacity, or external advising. The results are consistent with models in which organizational form proxies for entrepreneurial sophistication or planning capacity (e.g., Hurst and Pugsley (2011)), while also pointing to a role for frictions and uncertainty in expectation formation across legal forms.

Figure 3: Average expectation error by sector



Notes: Panels A and B show average expectation error (δ_{it}) by sector.

4.2.2 Sectoral heterogeneity

Figure 3 and Table A-1 in Appendix A show substantial sectoral variation in expectation errors, reflecting differences in sector-specific planning environments. Manufacturing, Health Care, and Information have the largest raw expectation errors, indicating systematic overprediction. These sectors often involve longer planning horizons, regulatory complexity, or demand uncertainty, which may lead entrepreneurs to overestimate initial employment. By contrast, Professional Services, Real Estate, Retail, and Accommodation and Food Services have smaller errors, possibly reflecting more standardized business models or lower initial staffing needs.

These differences largely persist among applicants with positive expected employment. Utilities and Manufacturing, for example, continue to show large errors, suggesting that realized hiring often falls short even when hiring is planned. This may reflect delays in capital deployment, permitting, or sales acquisition in execution-heavy sectors. Overall, the patterns are consistent with theories in which industry characteristics shape entrepreneurial forecasts (e.g., Wu and Knott (2006); Bloom (2009); Hall and Woodward (2010)).

4.2.3 High-tech business applications

As an example of finer industry-level heterogeneity, Table 4 reports employment expectations and errors for applicants intending to operate in high-tech industries. High-tech industries are defined by the concentration of Science, Technology, Engineering, and Math (STEM) employment following

Goldschlag and Miranda (2020) and Hecker (2005).[19]

High-tech applicants report slightly higher expected but lower realized employment than non–high-tech applicants, producing larger mean errors. Among applicants with positive expected employment, high-tech firms also overpredict more: their average error is 2.9 employees, compared with 2.4 for non–high-tech firms.

Table 4: Summary statistics for expectation errors by high-tech industry status

	All applications			Positive expectations ($\tilde{\mu}_{it} > 0$)		
High-tech status	$\tilde{\mu}_{it}$	m_{it}	δ_{it}	$\tilde{\mu}_{it}$	m_{it}	δ_{it}
No	0.535	0.395	0.141	4.280	1.863	2.417
	(0.001)	(0.001)	(0.001)	(0.006)	(0.005)	(0.006)
Yes	0.575	0.343	0.232	4.193	1.272	2.921
	(0.004)	(0.003)	(0.005)	(0.028)	(0.016)	(0.028)

Notes: Entries are sample means by high-tech industry status. "Positive expectations" refers to applications that report positive expected maximum employment. Standard errors are in parentheses. $\delta_{it} = \tilde{\mu}_{it} - m_{it}$.

Overall, employment expectations are less well calibrated in high-tech industries. This is consistent with theories emphasizing greater volatility and information frictions among technology-oriented startups (e.g., Hall and Woodward (2010)), suggesting that high-tech entrepreneurs may form expectations under greater uncertainty or optimism.

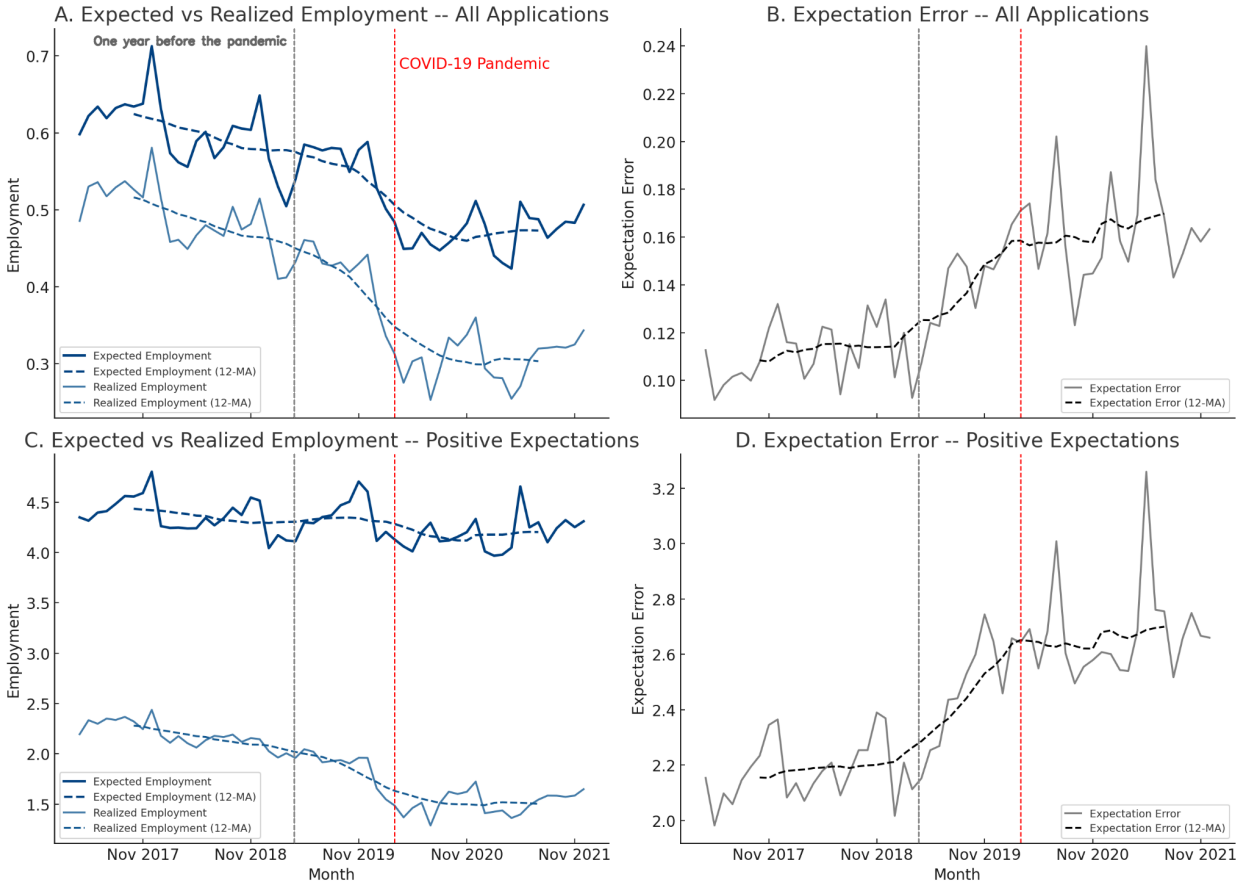
4.3 The evolution of expectation errors around the COVID-19 Pandemic

We next examine the time-series evolution of expectations, realizations, and errors around the COVID-19 shock. Figure 4 plots monthly average expected and realized employment (Panels A, C) and average expectation errors (Panels B, D), while Figure 5 shows the standard deviation and coefficient of variation (CV) of errors. Top panels use all applications; bottom panels restrict to those with positive expected employment.

In both samples, expected employment exceeds realized employment, and both decline between 2017 and 2019, consistent with longer-run declines in startup size. A clearer divergence begins

[19]The 4-digit NAICS industries classified as high-tech are: 3341 (Computer and Peripheral Equipment Manufacturing), 3342 (Communications Equipment Manufacturing), 3344 (Semiconductor and Other Electronic Component Manufacturing), 3345 (Navigational, Measuring, Electromedical, and Control Instruments Manufacturing), 3364 (Aerospace Product and Parts Manufacturing), 5112 (Software Publishers), 5182 (Data Processing, Hosting, and Related Services), 5191 (Other Information Services), 5413 (Architectural, Engineering, and Related Services), 5415 (Computer Systems Design and Related Services), and 5417 (Scientific Research and Development Services). See also Business Dynamics Statistics-High Tech for details.

Figure 4: The evolution of average expected and realized employment, and expectation error

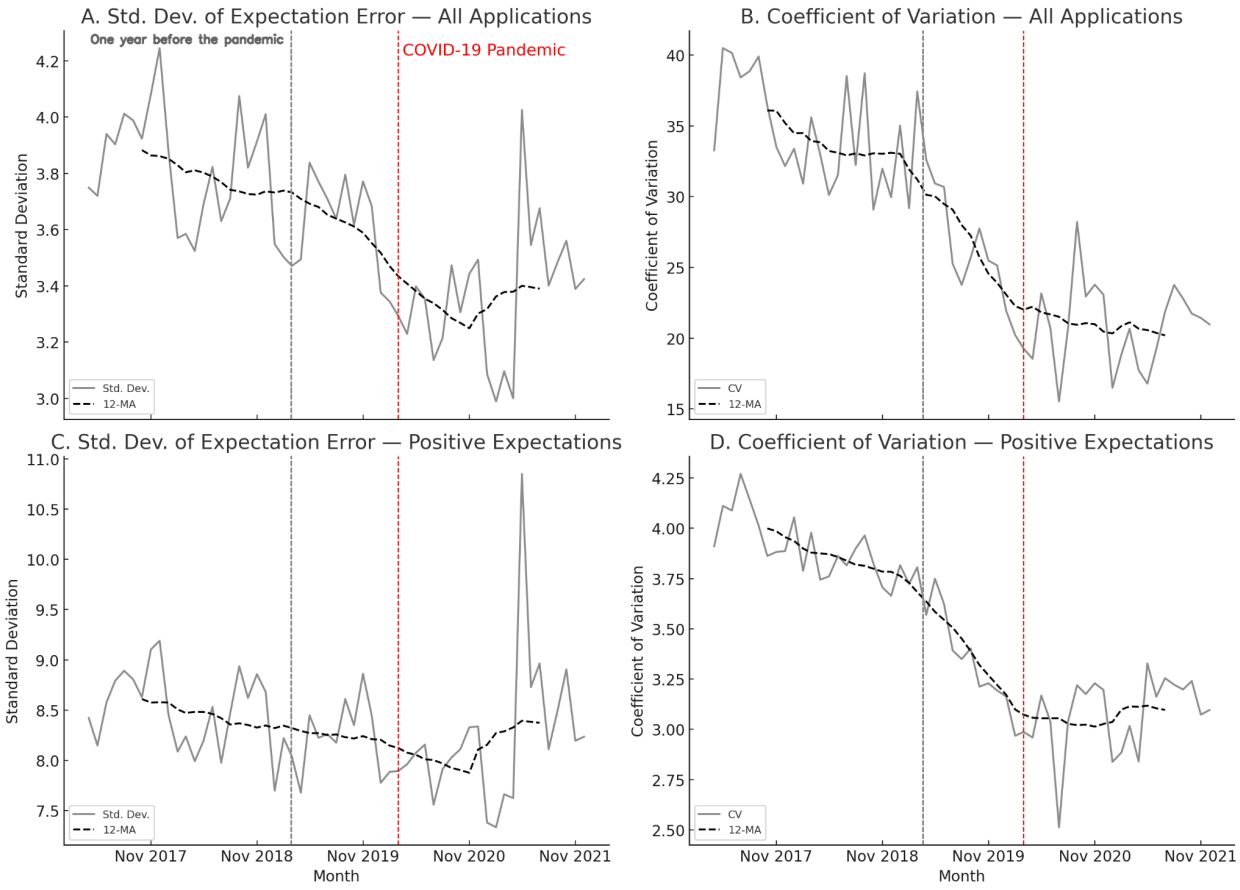


Notes: Dashed curves are 12-month centered moving averages (12-MA). Red vertical line indicates March 2020; grey vertical line indicates March 2019.

around March 2019, when the 12-month realization window for new applications starts overlapping with the pandemic onset in March 2020. Realized outcomes for applications submitted from that point onward can therefore be affected by the pandemic. Expectations decline less sharply than realizations during 2019–2020, increasing overestimation. As the pandemic approaches, both fall more rapidly, but after March 2020 expectations level off sooner while realizations continue declining before stabilizing, leaving a persistent post-COVID gap and elevated errors.

Figure 5 shows that the standard deviation and CV of expectation errors decline from 2019 through early 2020, with less compression in the positive-expectations sample (Panel C). Part of this convergence reflects a surge in applications with zero expected and realized employment, increasing the mass at zero error. By late 2020 or early 2021, dispersion rises again, especially in the standard deviation and more strongly among positive-expectation applications. Thus, the initial

Figure 5: The evolution of the standard deviation and coefficient variation of the expectation error



Notes: Dashed curves are 12-month centered moving averages (12-MA). Red vertical line = March 2020; grey vertical line = March 2019.

pandemic shock compressed error dispersion as expectations and realizations fell, but heterogeneity re-emerged during the recovery, consistent with differences in local and sectoral exposure.

4.4 Are expectations unbiased and efficient?

To formally assess forecast bias and efficiency, we estimate the standard Mincer–Zarnowitz regression (Mincer and Zarnowitz, 1969):[20]

$$m_{it} = \alpha + \beta \tilde{\mu}_{it} + u_{it}. \tag{9}$$

The joint null hypothesis of $\alpha = 0$ and $\beta = 1$ corresponds to unbiased and efficient forecasts.

[20] See also Chapter 3 in Bachmann, Topa, and van der Klaauw, 2022.

As a complementary diagnostic, we also estimate

$$\delta_{it} = \tilde{\mu}_{it} - m_{it} = \alpha^* + \beta^* \tilde{\mu}_{it} + u^*_{it}, \tag{10}$$

where $\alpha^* = -\alpha$, $\beta^* = 1 - \beta$, and $u^*_{it} = -u_{it}$. This algebraically equivalent specification facilitates interpretation of unconditional bias (α^*) and forecast-dependent errors (β^*).

Table 5: Mincer-Zarnowitz test of rational expectations

	All Applications		Positive Exp. ($\tilde{\mu}_{it} > 0$)	
	m_{it}	δ_{it}	m_{it}	δ_{it}
$\tilde{\mu}_{it}$	.288***	.712***	.261***	.739***
	(0.002)	(0.002)	(0.002)	(0.002)
const.	.235***	-.235***	.719***	-.719***
	(0.001)	(0.001)	(0.008)	(0.008)
R^2	0.091	0.381	0.103	0.478
N	16,820,000	16,820,000	2,082,000	2,082,000
F-statistic	68580		171600	
p-value	0.000		0.000	

Notes: Robust standard errors are in parentheses. Stars denote statistical significance: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. F-statistic pertains to the joint hypothesis $H_0 : \alpha = 0, \beta = 1$. N and F-statistic values are rounded for disclosure avoidance.

Table 5 reports both specifications. In the levels regressions, slope estimates are far below one (0.29 and 0.26) and intercepts are significantly positive, rejecting the joint null $\alpha = 0$ and $\beta = 1$. Correspondingly, the error regressions yield slopes of 0.71 and 0.74 and intercepts significantly different from zero, indicating that errors vary systematically with forecast levels and exhibit unconditional bias.

These results reject the unbiasedness and efficiency of reported expectations in the Mincer–Zarnowitz sense, but do not by themselves imply irrational expectation formation. The Mincer–Zarnowitz restrictions rely on additional assumptions, including that reported forecasts accurately measure applicants' conditional expectations and correspond to the realized outcome being evaluated. Measurement or reporting error, and differences between the reported forecast and the conditional-mean object assumed by the test, can contribute to departures from $\alpha = 0$ and $\beta = 1$ without necessarily implying irrational behavior. We therefore interpret the results as evidence of systematic departures from the Mincer–Zarnowitz null hypothesis rather than as a test of individual rationality. We next examine whether forecast errors are systematically related to observable information available at filing.

4.5 Regression analysis of expectation errors

To assess conditional forecast efficiency, we examine whether forecast errors are related to information available at application:

$$\delta_{it} = \alpha + \beta\tilde{\mu}_{it} + \gamma' z_{it} + q_t + \iota_{j(i)} + \xi_{s(i)} + u_{it}, \tag{11}$$

where z_{it} includes observable application characteristics and local economic conditions at filing. The specification also includes year–quarter (q_t), four-digit NAICS industry ($\iota_{j(i)}$), and state ($\xi_{s(i)}$) fixed effects.

Because δ_{it} mechanically contains $\tilde{\mu}_{it}$, we condition on the forecast level so that γ captures associations between other filing-time information and forecast errors. Under the benchmark that expectations efficiently incorporate the available information set, these observables should not systematically predict errors. A significant element of γ therefore indicates a departure from conditional forecast efficiency under this benchmark. Local demographic, economic, and business conditions are included because they may shape expected demand, costs, and post-entry growth and approximate information available to applicants at filing.

4.5.1 Covariates

For the covariates z_{it}, we include four blocks capturing application attributes and local conditions that proxy both the information available to founders and the environment shaping expectations and realizations. The analysis is descriptive, not causal; unobserved traits (e.g., ambition, risk tolerance) may confound associations. To capture information "available at filing", we use pre-application or contemporaneous/lagged local measures, in addition to application characteristics. Appendix Table A-2 details variable construction.

Application attributes. Legal form and prior experience proxy for managerial sophistication and resource access. Corporations and LLCs face greater reporting obligations and often employ professional advisers; sole proprietorships and informal partnerships are typically less formal.[21] We include dummies for legal form (non-LLC partnership, non-LLC corporation, non-LLC other, and LLC, with non-LLC sole proprietorship as reference) and an indicator for prior EIN issuance to the applicant: a potential proxy for experience, regulatory familiarity, and serial entrepreneurship.

[21] See U.S. Small Business Administration (2025).

Prior experience may be correlated with better information about a market or industry (Lafontaine and Shaw (2016)) but may also reflect persistent overconfidence.

Local demographic conditions. Using ACS data as in Dinlersoz et al. (2023), we relate errors to tract-level demographics. Higher college-educated share proxies richer human-capital pools and knowledge spillovers (Moretti, 2004a); non-white share captures entrepreneurial composition, demand shifters, and potential constraints (e.g., financing or hiring frictions) (Fairlie, 2012); median age (log) proxies local experience and life-cycle demand. All measures use a two-year lag to ensure availability at filing and mitigate simultaneity concerns.[22]

Local market structure. Market structure shapes competition and stability, and thus may affect forecast errors. In imperfect competition models, larger markets feature lower markups, lower firm density, higher average firm size, and greater turnover,[23] making outcomes harder to predict and potentially biasing expectations. Low density may also weaken agglomeration benefits and information flows. Higher concentration can grant incumbents market power that entrants underestimate, yet it may also stabilize markets and improve forecasting accuracy; Datta, Iskandar-Datta, and Sharma (2011) show concentrated industries exhibit lower volatility and better information efficiency. To capture market structure prior to application, we construct three proxies from LBD (lagged one year): (i) log firm density (firms per 1,000 population in county–industry (2-digit NAICS)); (ii) the DHS transform of county–industry average employment (local average firm size),[24] and (iii) the top-four-firm employment share.

Local dispersion in firm outcomes. Greater cross-sectional variability can hinder accurate expectations (Bachmann and Bayer, 2014) and foster overconfidence (Wu and Knott, 2006). We compute (lagged one year, LBD) at the county–industry level: (i) the coefficient of variation of firm employment (scale dispersion) and (ii) the coefficient of variation of labor productivity (revenue per employee), capturing heterogeneity in technology, prices, and efficiency.

Local shocks. To gauge contemporaneous shocks, we include an industry–county exposure to

[22] Using a one-year lag for 2017–2020 yields similar results given high persistence; see Dinlersoz et al. (2023).

[23] See Campbell and Hopenhayn (2005), Dinlersoz (2004), Nocke (2006), Asplund and Nocke (2006).

[24] Formally, $DHS(\overline{Emp})_{j,c,t-1} = \frac{\overline{Emp}_{j,c,t-1} - \overline{Emp}}{0.5(\overline{Emp}_{j,c,t-1} + \overline{Emp})}$, where j is 2-digit NAICS, c county, t application year, $\overline{Emp}_{j,c,t-1}$ the cell average, and $\overline{Emp}$ the overall average of firm employment.

COVID-19:

$$\text{COVIDShock}_{i,t} \;=\; (1 - \text{WFHshare}_{j(i)}) \times \frac{\text{NewCases}_{c(i),t}}{\text{Pop}_{c(i),2019}},$$

where $(1 - \text{WFHshare}_{j(i)})$ (from Dingel and Neiman 2020) measures the share of non-teleworkable jobs in industry $j(i)$, and the second term is the per-capita new cases in county $c(i)$ and quarter t. Higher values indicate less teleworkable industries in harder-hit areas. Following Papanikolaou and Schmidt (2022), the negative association between $(1 - \text{WFHshare}_{j(i)})$ and subsequent industry employment growth supports interpreting this as adverse local conditions at filing. $\text{COVIDShock}_{i,t}$ enables tests of whether forecasting errors vary with exposure to an adverse shock.[25] We also control for prior-quarter county GDP growth to separate local exposure from broader macro fluctuations.

Together, these covariates span: (i) application-level information quality, (ii) local market competition and agglomeration, (iii) cross-sectional heterogeneity in outcomes, and (iv) contemporaneous local shocks, enabling a descriptive assessment of how each correlates with entrepreneurial expectations, in line with work on business formation and managerial forecasting.

4.5.2 Results

Table 6 reports results for the full sample and the subsample with positive expected employment. In both samples, forecast errors are significantly related to expected employment and several observable characteristics at filing, indicating systematic variation in errors across applicants and local conditions. The coefficient on expected employment, $\tilde{\mu}_{it}$, is stable at about 0.75 across samples, implying that realized employment increases substantially less than one-for-one with expected employment and that overprediction rises with the forecast level.

Corporations and LLCs are associated with less overprediction, consistent with more formal businesses having greater planning capacity or access to information. A higher local college share is also associated with less overprediction, while a higher non-white share is associated with more, potentially reflecting differences in planning resources, information, or networks.

Local market conditions are also systematically related to forecast errors. Greater firm density and employment concentration are associated with less overprediction, while greater dispersion in

[25] The COVID shock can affect both the level of activity and uncertainty; moreover, uncertainty itself can operate like a level (aggregate-demand) shock (see Leduc and Liu (2016)).

Table 6: OLS regressions for expectation error, δ_{it}

Variable	All Applications	Pos. Exp. ($\tilde{\mu}_{it} > 0$)
$\tilde{\mu}_{it}$	0.746***	0.765***
	(0.002)	(0.002)
partnership	-0.065***	-0.227***
	(0.004)	(0.030)
corporation	-0.505***	-1.195***
	(0.003)	(0.013)
llc	-0.213***	-0.794***
	(0.001)	(0.010)
other	-0.088***	-0.140***
	(0.005)	(0.030)
prior EIN	0.040	0.286
	(0.048)	(0.156)
college share	-0.334***	-1.097***
	(0.007)	(0.040)
nonwhite share	0.225***	1.112***
	(0.003)	(0.020)
ln(median age)	0.011*	0.039
	(0.005)	(0.029)
ln(firm density)	-0.045***	-0.145***
	(0.002)	(0.012)
dhs(avg. firm emp.)	-0.036***	-0.151***
	(0.002)	(0.012)
top4 emp. share	-0.087***	-0.224***
	(0.006)	(0.036)
cv(firm emp.)	0.010***	0.032***
	(0.000)	(0.002)
cv(labor prod.)	0.003	0.019
	(0.002)	(0.013)
local gdp growth	0.024	0.260**
	(0.018)	(0.088)
local covid-shock	0.025***	0.066***
	(0.001)	(0.009)
const.	-0.416***	-1.341***
	(0.023)	(0.135)
year-quarter FE	Y	Y
industry FE (4-digit NAICS)	Y	Y
state FE	Y	Y
R^2	0.403	0.502
N	14,290,000	1,786,000

Notes: The omitted category for legal form of organization is sole proprietorship. Robust standard errors are in parentheses. Stars denote statistical significance: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. N values are rounded for disclosure avoidance. The difference in the number of observations (N) between this table and Table 5 is attributable to missing values in some covariates.

firm employment and productivity is associated with more. These patterns are consistent with denser or more stable business environments facilitating more accurate forecasts, and greater heterogeneity in local outcomes increasing uncertainty.

Greater exposure to the COVID-19 shock is associated with more overprediction, whereas GDP growth has no consistent relationship with forecast errors. This pattern is consistent with the descriptive evidence in Section 4.3 that adverse pandemic conditions widened the gap between expected and realized employment. Model fit is higher for applications with positive expected employment ($R^2 = 0.50$ versus 0.40), indicating that the specification accounts for a larger share of variation in forecast errors among applicants planning to hire.

4.6 Discrete entry choice and conditional employment outcome

The preceding OLS regressions treat zero employment as a realized outcome, which is appropriate because applications that do not become employers generate no employer employment during the observation window. Nevertheless, the mass at zero arises from the discrete employer-entry margin, while positive employment reflects the scale of activity conditional on entry. A single linear equation may therefore obscure differences between these two margins.

We use two complementary approaches. Our primary specification is a two-part model that directly estimates the probability of employer entry and the conditional mean of realized employment among entrants. We then use a Heckman selection model as a latent Gaussian benchmark that explicitly models correlation between unobserved determinants of entry and latent employment.

The two-part model specifies

$$\Pr(e_{it} = 1 \mid x_{it}^E) = \Phi\big(x_{it}^{E\prime}\psi^E\big), \qquad \mathbb{E}[m_{it} \mid e_{it} = 1, x_{it}^M] = x_{it}^{M\prime}\psi^M, \tag{12}$$

where x_{it}^E contains predictors of employer entry and x_{it}^M contains predictors of employment conditional on entry. Both vectors include reported expected employment, $\tilde{\mu}_{it}$, along with the relevant application characteristics, local conditions, and fixed effects.

The second equation is a linear conditional-mean specification for realized employment among entrants. It should not be interpreted as a literal distributional model for a nonnegative count outcome. We estimate the same identity-link conditional mean using Gaussian and Gamma GLMs. The Gaussian specification provides the OLS conditional-mean benchmark, while the Gamma spec-

ification allows a variance function more suited to positive, right-skewed employment. The Gamma identity-link specification requires positive fitted conditional means, but neither specification requires a logarithmic transformation of employment.

The zero-inclusive conditional mean can then be written as

$$\mathbb{E}\big[m_{it} \mid x_{it}^{E}, x_{it}^{M}\big] = \Pr\big(e_{it} = 1 \mid x_{it}^{E}\big)\ \mathbb{E}\big[m_{it} \mid e_{it} = 1, x_{it}^{M}\big]\,. \tag{13}$$

This factorization describes realized outcomes and does not require a latent employment equation to be defined for nonentrants. Correlation between the unobserved determinants of employer entry and post-entry employment is reflected in the selected conditional distribution of employment among entrants.

The Heckman model instead posits a latent linear employment equation:

$$e_{it} = \mathbf{1}\left\{x_{it}^{E\prime}\psi^{E} + u_{it}^{E} > 0\right\},\ m_{it}^{*} = x_{it}^{M\prime}\psi^{M} + u_{it}^{M},\ \begin{pmatrix} u_{it}^{E} \\ u_{it}^{M} \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho_H\sigma_H \\ \rho_H\sigma_H & \sigma_H^2 \end{pmatrix}\right). \tag{14}$$

The latent outcome m_{it}^{*} is observed as employment only when $e_{it} = 1$.

As in the Gaussian second part of the two-part model, m_{it}^{*} is a continuous linear approximation to employment in levels. Its normal distribution is used for selection correction and is not intended to impose the literal support of observed employment. The Heckman model therefore requires stronger functional-form and distributional assumptions than the two-part model.

The two-part and Heckman models answer different questions. The two-part employment equation estimates the mean employment of the selected population of actual entrants. The Heckman outcome equation instead represents a latent employment relationship for the broader applicant population and adjusts for selection into employer entry. The theoretical model in Section 5 corresponds most closely to this latent-selection interpretation.

Although the Heckman model can be identified through functional-form assumptions, identification is stronger with an exclusion restriction: a variable contained in x_{it}^{E} but excluded from x_{it}^{M} that affects entry without directly affecting post-entry employment. To motivate such a variable, write latent entry value as $V_{it} = W_{it} - K_{it}$, where W_{it} is the gross value of operating and K_{it} is a sunk entry cost. Variables that shift K_{it} without directly affecting operating value or scale can shift employer entry while leaving initial employment unchanged conditional on entry and controls.

We construct an exclusion restriction using procedural entry costs from Arizona State University's Doing Business in North America (DBNA) database. The DBNA reports the time and monetary costs of legal procedures required to establish a domestic limited liability company (LLC), including name registration, formation filings, publication requirements, and state and local registration. These geographically varying costs are natural shifters of entry because they pertain to business formation rather than operating scale.[26] Because the DBNA measures apply specifically to LLC formation and cover 83 relatively large U.S. cities spanning all 50 states, the Heckman analysis is restricted to LLC applications in these locations, for which our microdata provide a large estimation sample.

4.6.1 Two-part models

The two-part model results in Table 7 provide a more nuanced view of how expectations relate to both employer entry and realized employment conditional on entry. The first part estimates a Probit model for entry, and the second estimates GLMs for realized employment using identity links with Gaussian and Gamma error distributions. The Gaussian specification corresponds to OLS, while the Gamma specification provides robustness to the skewed distribution of realized employment conditional on entry.

Across specifications, expected maximum employment ($\tilde{\mu}_{it}$) significantly predicts both entry and post-entry employment. In the full sample (Panel A) with all covariates, the Probit coefficient is 0.035, while the GLM coefficients range from 0.503 (Gamma) to 0.675 (Gaussian), implying that realized employment rises with expectations but less than proportionally. In the positive-expectations sample (Panel B), expected employment only weakly predicts entry, indicating substantial uncertainty in the entry process. Conditional on entry, however, the GLM coefficients in specifications with all covariates rise to 0.766–0.887, suggesting closer alignment between expectations and realizations among applicants who anticipated hiring.[27]

The results help reconcile entrants' average tendency to underpredict employment with the

[26] The procedures include: (1) reserving or registering the LLC name, (2) choosing or assigning a registered agent, (3) filing articles of incorporation, organization, or formation, (4) completing state LLC publication requirements, (5) filing an initial statement of information, (6) creating an LLC operating agreement, (7) obtaining a state identification number, and (8) fulfilling additional county- or city-level requirements. See the DBNA website for methodological details.

[27] Column (12) omits local GDP growth and the local COVID-19 shock because the MLE fails to converge when they are included.

positive relationship between expected and realized employment. Applicants with higher expectations tend to hire more, but realized employment rises less than proportionally, so forecast errors increase with expected employment. Thus, underprediction can prevail among entrants even as overprediction becomes more pronounced at higher expectation levels – see Panel B of Figure 2.

Table 7: Two-part model estimation for actual maximum employment, m_{it} — All applications (left) vs. Positive expectations $\left(\tilde{\mu}_{it} > 0\right)$ (right)

	A. All applications						B. Positive expectations ($\tilde{\mu}_{it} > 0$)					
	Probit		Gaussian		Gamma		Probit		Gaussian		Gamma	
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
$\tilde{\mu}_{it}$	0.049***	0.035***	0.787***	0.675***	0.622***	0.503***	0.006***	0.005***	0.845***	0.766***	1.072***	0.887***
	(0.000)	(0.000)	(0.004)	(0.004)	(0.002)	(0.002)	(0.000)	(0.000)	(0.005)	(0.005)	(0.004)	(0.004)
partnership		0.269***		2.038***		1.337***		-0.115***		1.346***		0.802***
		(0.005)		(0.078)		(0.062)		(0.009)		(0.106)		(0.062)
corporation		1.091***		1.689***		0.751***		0.416***		1.604***		0.684***
		(0.002)		(0.027)		(0.016)		(0.005)		(0.035)		(0.019)
llc		0.579***		2.050***		0.896***		0.106***		1.630***		0.716***
		(0.002)		(0.026)		(0.016)		(0.004)		(0.035)		(0.018)
other		0.474***		1.668***		0.956***		-0.139***		0.863***		0.469***
		(0.005)		(0.090)		(0.074)		(0.011)		(0.130)		(0.089)
prior EIN		0.197***		-0.577*		-0.212		-0.132**		-0.367		-0.351**
		(0.024)		(0.244)		(0.163)		(0.041)		(0.464)		(0.133)
college share		0.245***		2.532***		1.273***		0.346***		1.601***		0.707***
		(0.004)		(0.074)		(0.049)		(0.009)		(0.103)		(0.059)
nonwhite share		-0.465***		-0.478***		0.048		-0.629***		-0.615***		-0.095*
		(0.003)		(0.048)		(0.035)		(0.006)		(0.071)		(0.040)
ln(med. age)		-0.037***		-0.343***		-0.211***		-0.021**		-0.188*		-0.132**
		(0.004)		(0.052)		(0.034)		(0.007)		(0.074)		(0.040)
ln(firm dens.)		0.141***		-0.258***		-0.113***		0.136***		-0.210***		-0.032
		(0.002)		(0.023)		(0.015)		(0.003)		(0.032)		(0.018)
dhs(emp.)		-0.031***		0.619***		0.373***		-0.036***		0.519***		0.220***
		(0.002)		(0.024)		(0.018)		(0.003)		(0.034)		(0.019)
top4 emp. share		0.277***		-0.888***		-0.203***		0.312***		-0.754***		-0.222***
		(0.005)		(0.068)		(0.045)		(0.009)		(0.092)		(0.051)
cv(emp.)		-0.011***		-0.033***		-0.030***		-0.012***		-0.034***		-0.018***
		(0.000)		(0.003)		(0.002)		(0.000)		(0.005)		(0.003)
cv(labor prod.)		-0.008***		0.035		0.011		-0.009***		0.002		-0.021
		(0.001)		(0.023)		(0.015)		(0.003)		(0.031)		(0.017)
gdp growth		0.003		-0.375*		-0.632***		-0.038*		-0.221		
		(0.010)		(0.171)		(0.144)		(0.018)		(0.226)		
covid-shock		-0.019***		-0.004		-0.006		-0.020***		-0.035		
		(0.001)		(0.017)		(0.011)		(0.002)		(0.024)		
const.	-1.521***	-3.096***	3.713***	0.734	3.924***	1.703***	-0.626***	-1.954***	2.643***	0.528	1.730***	0.836*
	(0.000)	(0.232)	(0.010)	(0.402)	(0.008)	(0.382)	(0.001)	(0.205)	(0.020)	(0.458)	(0.013)	(0.400)
year-quarter FE		Y		Y		Y		Y		Y		Y
industry FE (4-digit NAICS)		Y		Y		Y		Y		Y		Y
state FE		Y		Y		Y		Y		Y		Y
Log Likelihood			-8335000	-6506000	-7194000	-5475000			-3245000	-2664000	-2727000	-2266000
N			16,820,000	14,290,000	16,820,000	14,290,000			2,082,000	1,786,000	2,082,000	1,837,000

Notes: Columns (1)–(2) and (7)–(8) estimate the Probit model for $\Pr(e_{it} = 1 \mid x_{it}^{E})$ in equation (12); the remaining columns estimate GLMs with identity links for $\mathbb{E}[m_{it} \mid e_{it} = 1, x_{it}^{M}]$, assuming Gaussian and Gamma error distributions. Within each specification, the first column includes only $\tilde{\mu}_{it}$ and the second adds the full set of controls and fixed effects. In column (12), the MLE including local GDP growth and the local COVID-19 shock failed to converge, so these controls are omitted. The omitted legal form category is sole proprietorship. Standard errors in parentheses. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. Log likelihood is reported for the GLM component. N and log likelihood values are rounded for disclosure avoidance.

Other covariates provide additional insights. In the full sample, relative to sole proprietors, LLCs, corporations, and partnerships are significantly more likely to enter and, conditional on entry, achieve higher employment. LLCs and corporations are associated with 1.7–2.1 more employees (Gaussian GLM), consistent with more formal organizational types pursuing more ambitious or

better-resourced plans.

Prior EIN holders are more likely to enter but show smaller or insignificant differences in post-entry size. In the positive-expectations sample, their association with entry is negative and conditional employment slightly lower, suggesting that prior experience does not necessarily translate into more expansive early-stage hiring.

A higher local college share is associated with both greater entry and realized employment, while a higher non-white share is associated with lower levels of both. Market structure effects are mixed: higher firm density and employment concentration are associated with more entry but lower post-entry size, whereas greater employment dispersion is associated with lower entry and lower conditional employment.

The COVID-19 shock is associated with slightly lower entry but not significantly with conditional employment. GDP growth effects are also mixed: in the full sample, growth is negatively associated with realized employment under both GLM specifications, while in the positive-expectations sample the coefficient is negative but insignificant.

Overall, the two-part results show that expectations strongly correlate with realized outcomes, although forecast errors remain substantial and systematic. Year-specific bivariate estimates in Table 8 show that the expectation–outcome relationship remains present from 2017–2021 but changes in magnitude. In the full sample, Probit and GLM coefficients on $\tilde{\mu}_{it}$ decline over time, possibly reflecting greater pandemic-era uncertainty. In the positive-expectations sample, Probit coefficients remain small but significant and GLM coefficients larger, though also declining. Thus, expectations remain predictive, but their association with realizations weakens after COVID.

4.6.2 Heckman models

Table 9 presents Heckman maximum likelihood estimates for LLC applicants. The selection equation includes *ltime*, the log of local procedural days required to establish an LLC, as an exclusion restriction.[28] In both samples, expected maximum employment ($\tilde{\mu}_{it}$) strongly predicts both entry and conditional realized employment. As in the two-part models, outcome-equation coefficients are well below one, indicating systematic overprediction: realized employment rises with expectations

[28] This variable is the logarithm of the estimated number of days required to complete LLC startup tasks in a given city; see DBNA website.

Table 8: Two-part model estimation by year

A. All Applications

Variable	2017	2018	2019	2020	2021
			Probit		
$\tilde{\mu}_{it}$	0.051***	0.052***	0.052***	0.047***	0.042***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
const.	-1.411***	-1.436***	-1.469***	-1.595***	-1.617***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
			GLM		
$\tilde{\mu}_{it}$	0.812***	0.806***	0.789***	0.768***	0.759***
	(0.010)	(0.009)	(0.009)	(0.009)	(0.009)
const.	3.931***	3.749***	3.693***	3.632***	3.626***
	(0.027)	(0.023)	(0.022)	(0.022)	(0.021)
Log Likelihood	-1202000	-1771000	-1680000	-1672000	-1991000
N	2,003,000	3,084,000	3,096,000	3,843,000	4,791,000

B. Positive Expectations ($\tilde{\mu}_{it} > 0$)

Variable	2017	2018	2019	2020	2021
			Probit		
$\tilde{\mu}_{it}$	0.006***	0.006***	0.007***	0.006***	0.006***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
const.	-0.478***	-0.514***	-0.570***	-0.731***	-0.763***
	(0.003)	(0.002)	(0.002)	(0.002)	(0.002)
			GLM		
$\tilde{\mu}_{it}$	0.873***	0.864***	0.852***	0.827***	0.813***
	(0.012)	(0.010)	(0.010)	(0.011)	(0.010)
const.	2.823***	2.713***	2.540***	2.547***	2.609***
	(0.050)	(0.043)	(0.042)	(0.045)	(0.042)
Log Likelihood	-506500	-730500	-663000	-609200	-727500
N	283,000	421,000	403,000	436,000	538,000

Notes: Panel A reports estimates for all applications; Panel B restricts to applications with positive expected employment ($\tilde{\mu}_{it} > 0$). Each panel includes the Probit model for $\Pr(e_{it} = 1 \mid x_{it}^{E})$ and a GLM with Gaussian specification for $\mathbb{E}[m_{it} \mid e_{it} = 1, x_{it}^{M}]$ in equation (12), with $\tilde{\mu}_{it}$ as the only regressor. Log likelihood is the total log likelihood of both stages. Standard errors in parentheses. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. N and Log Likelihood values are rounded for disclosure avoidance.

but less than one-for-one.

With all covariates included, the coefficients on $\tilde{\mu}_{it}$ in the selection equation are 0.033 in the full sample and 0.007 in the positive-expectations subsample, reflecting less variation in entry intent when conditioning on positive expectations. Thus, expectations remain predictive of outcomes in the Heckman framework, consistent with the two-part models.

The exclusion restriction *ltime* is negative and significant in the full sample, suggesting that greater procedural burdens deter entry, but negative and statistically insignificant in the positive-expectations sample. Administrative burdens may therefore matter more for marginal applicants than for those with clearer hiring plans.

The selection correction indicates negative selection in both samples. The estimated correlation between unobservables ($\hat{\rho}_H$) is negative and significant, as is the implied selection term ($\hat{\rho}_H \hat{\sigma}_H$). Thus, unobserved characteristics that increase the likelihood of entry are associated with lower conditional realized employment. Several mechanisms could generate such negative selection. Applicants with lower opportunity costs or fewer outside options may enter more readily but operate at smaller scale, while those with more ambitious plans may face greater entry barriers, consistent with the distinction between subsistence and transformational entrepreneurs in Schoar (2010). These patterns reinforce the importance of ex-ante heterogeneity in early-stage employment outcomes. Covariate coefficients also largely align with the two-part results for LLCs (Tables A-3 and A-4), suggesting robustness to alternative treatments of zero employment outcomes.

The Heckman specification nevertheless relies on strong assumptions. The exclusion restriction must be unrelated to unobserved determinants of post-entry outcomes, which may fail if *ltime* also captures broader planning or regulatory quality. Joint normality and functional-form assumptions may also affect the estimated selection effects. Still, the results are broadly consistent with the two-part models: expectations are informative but imperfect and overstate realized scale.

4.7 Expectation errors and near-term outcomes

Table 10 reports OLS estimates relating initial expectation errors, δ_{it}, to two outcomes measured three years (twelve quarters) after application: (i) an indicator for positive employment and (ii) employment level. Applications that never become employer businesses or that enter and exit before the end of year three are assigned zero employment. The sample includes applications from

Table 9: Heckman maximum-likelihood estimation for the sample of LLC applications

	A. Selection Equation				B. Outcome Equation			
	All		Pos. Exp. ($\tilde{\mu}_{it} > 0$)		All		Pos. Exp. ($\tilde{\mu}_{it} > 0$)	
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\tilde{\mu}_{it}$	0.043***	0.033***	0.013***	0.007***	0.634***	0.563***	0.765***	0.709***
	(0.001)	(0.001)	(0.001)	(0.000)	(0.016)	(0.015)	(0.016)	(0.016)
prior EIN		0.378***		-0.707***		-0.898		-2.306
		(0.088)		(0.204)		(1.071)		(1.675)
college share		0.267***		0.416***		2.730***		2.010***
		(0.013)		(0.027)		(0.274)		(0.415)
nonwhite share		-0.499***		-0.680***		1.287***		0.256
		(0.010)		(0.021)		(0.213)		(0.333)
ln(med. age)		-0.067***		-0.033		0.075		0.106
		(0.012)		(0.023)		(0.228)		(0.346)
ln(firm dens.)		0.119***		0.118***		-1.012***		-0.920***
		(0.007)		(0.015)		(0.141)		(0.225)
dhs(emp.)		0.010		-0.007		0.056		0.096
		(0.007)		(0.015)		(0.139)		(0.208)
top4 emp. share		0.252***		0.401***		-3.004***		-2.427**
		(0.032)		(0.066)		(0.607)		(0.905)
cv(emp.)		-0.009***		-0.014***		0.094***		0.033
		(0.001)		(0.002)		(0.020)		(0.031)
cv(labor prod.)		-0.006		-0.028*		0.157		0.220
		(0.006)		(0.011)		(0.124)		(0.175)
gdp growth		0.006		-0.043		0.637		0.509
		(0.026)		(0.048)		(0.527)		(0.705)
covid shock		-0.023***		-0.015*		0.023		0.020
		(0.004)		(0.007)		(0.066)		(0.103)
ltime	-0.006***	-0.011***	-0.005	-0.010				
	(0.002)	(0.003)	(0.004)	(0.006)				
const.	-1.711***	-1.706***	-0.956***	-5.825	12.240***	2.560	8.234***	-1.579
	(0.003)	(0.488)	(0.006)	(4.705)	(0.497)	(2.068)	(0.857)	(2.548)
year-quarter FE		Y		Y		Y		Y
industry FE (4-digit NAICS)		Y		Y		Y		Y
state FE		Y		Y		Y		Y
Log Likelihood					-667,000	-554,500	-228,400	-193,800
N					1,870,000	1,630,000	198,000	178,000
$\hat{\rho}_H$					-0.356***	-0.352***	-0.340***	-0.149***
$\hat{\sigma}_H$					10.602***	9.667***	10.227***	9.282***
$\hat{\rho}_H\hat{\sigma}_H$					-3.779***	-3.398***	-3.482***	-1.383***

Notes: Heckman MLE estimation results for equation (14). Panel A reports the selection equation, a Probit model for $\Pr(e_{it} = 1)$. Panel B reports the outcome equation for latent employment, m^*_{it}, estimated jointly with Panel A. $\hat{\rho}_H$ is the estimated correlation between u^E_{it} and u^M_{it}, $\hat{\sigma}_H$ the estimated standard deviation of u^M_{it}, and $\hat{\rho}_H\hat{\sigma}_H$ the coefficient on the selection term. Standard errors in parentheses. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. N and log likelihood values are rounded for disclosure avoidance.

2017–2019 with observable three-year outcomes.

All regressions control for realized initial employment, m_{it}, so comparisons hold initial size fixed. Consistent with canonical firm-dynamics models (Jovanovic, 1982; Hopenhayn, 1992), initial realized employment strongly predicts both three-year employer status and employment. Forecast errors, however, remain predictive even conditional on initial size.

In the full sample, greater initial overestimation is positively associated with both employer status and employment three years later. Among applicants with positive initial employment expectations, overestimation is negatively associated with employer status at the end of year three but positively associated with employment levels.

These findings go beyond the implications of canonical rational-expectations frameworks. In Jovanovic (1982), which abstracts from selection at the entry margin, entrants share a common prior and posterior beliefs depend on realized productivity; once early outcomes are controlled, forecast errors should not predict subsequent performance. A similar implication holds in Hopenhayn (1992).[29] With heterogeneous priors containing additional information, however, forecast errors may retain predictive content even after conditioning on initial outcomes.

More broadly, rational-expectations models with *ex-ante* heterogeneity and entry selection (e.g., Lucas, 1978; Kihlstrom and Laffont, 1979; Jovanovic, 1994) allow ability, risk preferences, or managerial skill to shape both entrepreneurial entry and firm size. Such models can generate associations between pre-entry characteristics and later outcomes, consistent with part of the evidence. However, they do not model subjective priors or explicit forecasts, and cannot directly account for systematic optimism or for forecast errors that remain predictive conditional on early outcomes.

Behavioral perspectives provide complementary explanations. Evidence on entrepreneurial optimism and overconfidence (e.g., Koellinger et al., 2007; Astebro et al., 2014) suggests that subjective beliefs may systematically depart from rational posteriors and affect subsequent outcomes. Optimism may reduce responsiveness to early negative signals or weaken credibility with financiers, employees, or customers, lowering the likelihood of remaining an employer. At the same time, ambitious expectations may encourage greater resource mobilization and lead to higher employment among surviving businesses.

[29] In Hopenhayn (1992), entrants have common priors, equal to the mean of the initial productivity distribution of entrants, and observe their initial productivity draw upon entry.

Table 10: Expectation errors and near-term (3-year) outcomes

	All Applications		Positive Exp. ($\tilde{\mu}_{it} > 0$)	
Variable	3-yr. pos. emp.	3-yr. emp.	3-yr. pos. emp.	3-yr. emp.
δ_{it}	0.001***	0.082***	-0.004***	0.085***
	(0.000)	(0.005)	(0.000)	(0.007)
m_{it}	0.025***	0.676***	0.017***	0.660***
	(0.000)	(0.007)	(0.000)	(0.009)
partnership	0.016***	0.104***	0.010**	0.537**
	(0.000)	(0.018)	(0.003)	(0.170)
corporation	0.119***	0.268***	0.145***	0.688***
	(0.000)	(0.009)	(0.002)	(0.028)
llc	0.049***	0.188***	0.080***	0.659***
	(0.000)	(0.004)	(0.002)	(0.023)
other	0.041***	0.026	0.039***	0.191
	(0.001)	(0.029)	(0.004)	(0.191)
prior EIN	0.011	-0.068	-0.032	-0.133
	(0.008)	(0.068)	(0.017)	(0.193)
college share	0.022***	0.319***	0.072***	0.939***
	(0.001)	(0.024)	(0.004)	(0.127)
nonwhite share	-0.034***	-0.111***	-0.099***	-0.510***
	(0.000)	(0.010)	(0.003)	(0.006)
ln(median age)	-0.001	-0.018	-0.002	-0.032
	(0.001)	(0.013)	(0.003)	(0.064)
ln(firm density)	0.014***	0.0006	0.022***	-0.065***
	(0.000)	(0.006)	(0.001)	(0.031)
dhs(avg. firm emp.)	-0.003***	0.064***	-0.004*	0.274***
	(0.000)	(0.007)	(0.001)	(0.038)
top4 emp. share	0.026***	0.013	0.059***	-0.035
	(0.001)	(0.018)	(0.004)	(0.099)
cv(firm emp.)	-0.001***	-0.008***	-0.003***	-0.022***
	(0.000)	(0.001)	(0.000)	(0.004)
cv(labor prod.)	-0.0007*	0.006	-0.001	0.007
	(0.000)	(0.006)	(0.001)	(0.030)
local gdp growth	0.002	0.052	-0.023	-0.018
	(0.006)	(0.116)	(0.026)	(0.513)
const.	0.110***	0.061	0.286***	-0.496
	(0.004)	(0.062)	(0.015)	(0.306)
year-quarter FE	Y	Y	Y	Y
industry FE (4-digit NAICS)	Y	Y	Y	Y
state FE	Y	Y	Y	Y
R^2	0.167	0.173	0.171	0.189
N	4,338,000	4,338,000	604,000	604,000

Notes: Each column shows a regression of the corresponding outcome on initial expectations and control variables. Columns (1) and (3) pertain to linear probability models (LPM) of 3-year employer status indicator (i.e. indicator of positive employment), and columns (2) and (4) are based on OLS models of 3-year employment level (including zero employment). Omitted category for legal form of organization is sole proprietorship. Standard errors in parentheses. *** *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. N values are rounded for disclosure avoidance.

5 Heterogeneous priors, learning, and selection before entry

The model extends the classic entrepreneurial learning framework of Jovanovic (1982), with each additional element introduced only as needed to explain three core findings: (i) in the Mincer–Zarnowitz regression (Table 9), the coefficient on expected first-year employment is in $(0, 1)$; (ii) the first-year expectation error is positively correlated with subsequent employment, conditional on realized first-year employment; and (iii) business applicants overestimate their first-year employment, but they underestimate it conditional on expected *and* actual entry.

5.1 Model setup

Throughout this section, we suppress the application period t and index periods relative to application: period 0 is the time of application, period 1 is the first-year horizon from t to $t + \tau$ in Section 2, and $k \geq 2$ indexes subsequent years. Thus $\tilde{\mu}_{i1}$, m_{i1}, e_{i1}, and δ_{i1} correspond to $\tilde{\mu}_{it}$, m_{it}, e_{it}, and δ_{it} in Sections 2–4, and m_{ik} denotes employment k years after application.

As in Jovanovic (1982), each potential entrant i has a latent business size governed by a random variable $\theta_i \sim \mathcal{N}(\bar{\theta}, \sigma_\theta^2)$. The latent size determines employer entry, e_{i1}, through a latent net-value index $V_{i1} = \theta_i + \eta_{i1}$ and latent first-year employment through $m_{i1}^* = \theta_i + \varepsilon_{i1}$, where $\eta_{i1} \sim \mathcal{N}(0, \sigma_\eta^2)$ and $\varepsilon_{i1} \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ with $\mathrm{Corr}(\eta_{i1}, \varepsilon_{i1}) = \rho_{\eta\varepsilon}$; realized first-year employment is $m_{i1} = e_{i1} m_{i1}^*$.[30] *Before* application, the potential entrant observes an informative signal $s_{i0} = \theta_i + \varepsilon_{i0}$, with $\varepsilon_{i0} \sim \mathcal{N}(0, \sigma_0^2)$. In addition, she holds an additive heterogeneous prior $\phi_i \sim \mathcal{N}(0, \sigma_\phi^2)$, independent of $(\theta_i, \varepsilon_{i0}, \eta_{i1}, \varepsilon_{i1})$, so that her prior is $\theta_i \sim \mathcal{N}(\bar{\theta} + \phi_i, \sigma_\theta^2)$. The heterogeneity ϕ_i can be interpreted as idiosyncratic optimism/pessimism or as subjective evaluation of common information; in either case, it is uninformative about the latent size. Given the prior mean, $\bar{\theta} + \phi_i$, and the observed signal s_{i0}, the subjective posterior mean (the agent's forecast of θ_i, and thus of m_{i1}^*) is

$$\tilde{\mu}_{i1}^* = \tilde{\mathbb{E}}[\theta_i \mid s_{i0}, \phi_i] = (1 - \kappa_0)(\bar{\theta} + \phi_i) + \kappa_0 s_{i0}, \qquad \kappa_0 = \frac{\sigma_\theta^2}{\sigma_\theta^2 + \sigma_0^2},$$

and the cross-sectional variance of forecasts is $\sigma_{\tilde{\mu}_1^*}^2 = \mathrm{Var}(\tilde{\mu}_{i1}^*) = \kappa_0 \sigma_\theta^2 + (1 - \kappa_0)^2 \sigma_\phi^2$.

The orthogonality of ϕ_i to θ_i is a normalization rather than a substantive restriction: Given

[30] Local market conditions set by incumbents do not depend on other applicants' beliefs and are part of θ_i. We abstract from interactions within the entering cohort, in which an applicant's realized size would depend on how many other applicants enter. Such interactions would break the identity $\beta_{\mathrm{MZ}} = \mathrm{Var}(\tilde{\mu}_1^{0*})/\mathrm{Var}(\tilde{\mu}_1^*)$ and, under substitutability, overstate the distortions in Section 5.3. We expect these interactions to be weak, since new entrants are small in employment relative to incumbents and have little effect on one another's outcomes.

$\text{Cov}(\theta_i, \phi_i) = h \neq 0$, one can write $\phi_i = (h/\sigma_\theta^2)(\theta_i - \bar{\theta}) + \phi_i^\perp$ with $\phi_i^\perp \perp \theta_i$, where the first component acts like additional signal precision and $\phi_i^\perp$ plays the role of the orthogonal shift. Since beliefs affect entry and employment only through θ_i, all observable moments depend on the forecast only through $\text{Cov}(\theta_i, \tilde{\mu}_{i1}^*)$ and $\text{Var}(\tilde{\mu}_{i1}^*)$, so the correlated and orthogonal specifications are observationally equivalent and h is not separately identified.[31]

Suppose that there is a one-time cost of an application (cost of exploration/idea formation), and similarly, of actual entry (costs of operational launch).[32] Under a monotone mapping from employment to net value, these costs induce thresholds c_A and c_E such that a potential entrant applies if $\tilde{\mu}_{i1}^* > c_A$, expects entry if $\tilde{\mu}_{i1}^* > c_E$, and enters if $V_{i1} > c_E$. We write $\tilde{a}_{i1} = \mathbf{1}\{\tilde{\mu}_{i1}^* > c_A\}$, $\tilde{e}_{i1} = \mathbf{1}\{\tilde{\mu}_{i1}^* > c_E\}$, and $e_{i1} = \mathbf{1}\{V_{i1} > c_E\}$, so that expected size is $\tilde{\mu}_{i1} = \tilde{e}_{i1}\tilde{\mu}_{i1}^*$. For $k \geq 2$, latent employment follows $m_{ik}^* = \theta_i + \varepsilon_{ik}$, with $\varepsilon_{ik} \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ *i.i.d.* across k and independent of all else, employer status is $e_{ik} = \mathbf{1}\{V_{ik} > c_E\}$ with $V_{ik} = \theta_i + \eta_{ik}$ and $\eta_{ik} \sim \mathcal{N}(0, \sigma_\eta^2)$ i.i.d. across k, $\text{Corr}(\eta_{ik}, \varepsilon_{ik}) = \rho_{\eta\varepsilon}$, and realized employment is $m_{ik} = e_{ik} m_{ik}^*$.

5.2 Main implications

The model yields the following implications, with proofs in Appendix C. The first requires both an informative signal, s_{i0}, and uninformative prior heterogeneity, ϕ_i; the second shows that the informative component carries information about persistent business quality beyond first-year employment; and the third shows how prior heterogeneity and selection generate overestimation among applicants and underestimation among entrants.

(i) Mincer-Zarnowitz slope $\in (0,1)$: In the regression of m_1^* on $\tilde{\mu}_1^*$, the population slope is

$$\beta_{\text{MZ}} = \frac{\kappa_0 \, \sigma_\theta^2}{\kappa_0 \sigma_\theta^2 + (1-\kappa_0)^2 \sigma_\phi^2} \in (0,1),$$

provided $\sigma_0^2, \sigma_\phi^2 \in (0, +\infty)$. Only the informative component co-moves with latent employment, while both the informative and heterogeneous-prior components contribute to forecast variance. Hence $\beta_{\text{MZ}} = 0$ without an informative signal and $\beta_{\text{MZ}} = 1$ without uninformative prior hetero-

[31] The correlated case covers information-processing biases such as overprecision, where the applicant overweights her signal: $\tilde{\mu}_{i1}^* = (1-\tilde{\kappa}_0)\bar{\theta} + \tilde{\kappa}_0 s_{i0}$ with $\tilde{\kappa}_0 > \kappa_0$. Rewriting yields $\phi_i = \frac{\tilde{\kappa}_0 - \kappa_0}{1-\kappa_0}(s_{i0} - \bar{\theta})$, so $\text{Cov}(\theta_i, \phi_i) = \frac{\tilde{\kappa}_0 - \kappa_0}{1-\kappa_0}\sigma_\theta^2 > 0$: favorable signals generate excessively high expectations, unfavorable signals excessively low ones. Such an applicant is observationally indistinguishable from a Bayesian with a more precise signal plus an orthogonal prior shift.

[32] While the application itself may have a nominal cost, costs in the application stage broadly include those associated with planning and preparation for the business, such as estimating demand, seeking financing, understanding relevant regulations, and socializing the business idea and obtaining advice.

geneity. The slope is unchanged by conditioning on applicants $A = \{\tilde{\mu}_1^* > c_A\}$ or on positive expected employment $B = \{\tilde{\mu}_1^* > c_E\}$; Appendix C.1 provides the proof. The Heckman regression (Table 9) recovers β_{MZ} by correcting for entry selection, yielding estimates strictly inside $(0, 1)$.

(ii) Predictive content of expectation errors: Consider the near-term regression in Table 10,

$$m_{ik} = \beta_0^{\mathcal{S}} + \beta_1^{\mathcal{S}} m_{i1} + \beta_2^{\mathcal{S}} \delta_{i1} + u_{ik}, \qquad k \geq 2, \tag{15}$$

which relates zero-inclusive employment k years after application to realized first-year employment m_{i1} and the reported expectation error $\delta_{i1} = \tilde{\mu}_{i1} - m_{i1}$, all observed in the data. It is estimated by OLS within each sample $\mathcal{S}$: all applicants; those with positive expected employment ($\tilde{\mu}_{i1} > 0$); those with positive first-year employment ($m_{i1} > 0$); and those with both. Since m_{i1} is a regressor, $\beta_2^{\mathcal{S}}$ equals the coefficient on the reported forecast $\tilde{\mu}_{i1}$: it measures whether expectations predict later outcomes beyond what first-year employment reveals.

Without an informative signal ($\sigma_0^2 = +\infty$, so $\kappa_0 = 0$), $\tilde{\mu}_{i1}^* = \bar{\theta} + \phi_i$, so the reported forecast depends only on ϕ_i, while realized entry and employment depend only on θ_i and shocks independent of ϕ_i. Since each sample restricts either the forecast or realized first-year employment, the forecast remains independent of (m_{i1}, m_{ik}) within each sample, and $\beta_2^{\mathcal{S}} = 0$ (Appendix C.2). Hence $\beta_2^{\mathcal{S}} \neq 0$ implies that expectations carry information about latent business size, provided ϕ_i does not directly affect realized outcomes. In Table 10, the coefficients for $k = 3$ are positive and significant in all four samples, evidence of an informative pre-application signal.

(iii) Overall overestimation and conditional underestimation: Because applications involve fixed costs, only agents whose expected latent first-year employment exceeds c_A apply. Suppressing the subscript i, the latent forecast error among applicants is

$$\mathbb{E}\left[\delta_1^* \,\middle|\, \tilde{\mu}_1^* > c_A\right] = \frac{(1-\kappa_0)^2\,\sigma_\phi^2}{\sigma_{\tilde{\mu}_1^*}}\,\lambda\!\left(\frac{c_A - \bar{\theta}}{\sigma_{\tilde{\mu}_1^*}}\right) \;>\; 0,$$

where $\lambda(\cdot) = \varphi(\cdot)/(1-\Phi(\cdot))$: selection on high $\tilde{\mu}_1^*$ selects partly on the nonpersistent component ϕ. Conditioning instead on actual entry truncates $V_1 = \theta + \eta_1$ from below, raising the conditional mean of θ and hence of m_1^*, so

$$\mathbb{E}[\delta_1^* \mid V_1 > c_E] = -\,\frac{(1-\kappa_0)\sigma_\theta^2 + \rho_{\eta\varepsilon}\sigma_\eta\sigma_\varepsilon}{\sqrt{\sigma_\theta^2 + \sigma_\eta^2}}\,\lambda\!\left(\frac{c_E - \bar{\theta}}{\sqrt{\sigma_\theta^2 + \sigma_\eta^2}}\right),$$

which is negative whenever $(1-\kappa_0)\sigma_\theta^2 + \rho_{\eta\varepsilon}\sigma_\eta\sigma_\varepsilon > 0$. Appendix C.3 gives a sufficient condition for

strict negativity when conditioning additionally on expected entry, $\tilde{\mu}_1^* > c_E$. Heterogeneous priors thus generate overestimation among applicants, while entry selection generates underestimation among those who both expect and achieve entry.

The reported error censors both sides, $\delta_1 = \delta_1^* - (1 - \tilde{e}_1)\tilde{\mu}_1^* + (1 - e_1)m_1^*$, so the observed error equals δ_1^* only on $\{\tilde{e}_1 = e_1 = 1\}$.

The common-prior baseline (Jovanovic (1982)) alone cannot jointly account for a Mincer–Zarnowitz slope strictly between zero and one, the incremental predictive content of forecast errors, and aggregate overestimation. Table 11 summarizes the role of each model ingredient. The combination of the informative signal and heterogeneous priors produces a Mincer–Zarnowitz slope in $(0, 1)$, the informative signal delivers predictive content for subsequent employment, and heterogeneous priors generate the contrast between aggregate overestimation and conditional underestimation. Put differently, the forecast combines signal and noise: its informative component accounts for the predictive content of expectation errors, while its uninformative component generates forecast distortions and selection effects.

Table 11: Implications of the model's elements

	Baseline	Model element		
	Jovanovic (1982)	Informative signal (s_0)	Heterogeneous prior (ϕ)	Both (s_0, ϕ)
Mincer–Zarnowitz slope in $(0, 1)$	N (not identified)	N ($= 1$)	N ($= 0$)	Y
$\beta_2 \neq 0$ in regression (15)	N (not identified)	Y	N	Y
Over-/underestimation (latent)	N	N	Y	Y

5.3 Entry distortions from heterogeneous priors

The distortions generated by the uninformative belief component ϕ occur at two margins: the *expected*-entry margin (who plans to hire) and the *actual*-entry margin (who becomes an employer). The relevant counterfactual belief is the rational, ϕ-free latent forecast

$$\tilde{\mu}_{i1}^{0*} \equiv (1 - \kappa_0)\bar{\theta} + \kappa_0 s_{i0}, \tag{16}$$

so that the reported counterfactual forecast is $\tilde{\mu}_{i1}^0 = \mathbf{1}\{\tilde{\mu}_{i1}^{0*} > c_E\}\,\tilde{\mu}_{i1}^{0*}$. The actual latent forecast can be expressed as $\tilde{\mu}_{i1}^* = \tilde{\mu}_{i1}^{0*} + (1 - \kappa_0)\phi_i$. An agent applies if $\tilde{\mu}_{i1}^* > c_A$, plans entry if $\tilde{\mu}_{i1}^* > c_E$,

and enters as an employer if $V_{i1} > c_E$.[33]

5.3.1 Definitions

Definition 1 (Excess and forgone expected entry[34])**.** *Excess expected entry rate is the likelihood that an agent expects to enter, but would not absent ϕ, and forgone expected entry rate pertains to the reverse case:*

$$p^{exc} = \Pr\left(\tilde{\mu}_1^{0*} \leq c_E,\ \tilde{\mu}_1^{*} > c_E\right), \tag{17}$$

$$p^{for} = \Pr\left(\tilde{\mu}_1^{0*} > c_E,\ \tilde{\mu}_1^{*} \leq c_E\right). \tag{18}$$

Net excess expected entry and gross expected entry distortion rates are then

$$p^{net} = p^{exc} - p^{for}, \qquad p^{dist} = p^{exc} + p^{for}. \tag{19}$$

Definition 2 (Excess and forgone actual entry)**.** *Excess actual entry rate is the likelihood that an agent enters only because of ϕ, and forgone actual entry rate is the likelihood that ϕ deters an application that would have resulted in entry:*

$$p_{act}^{exc} = \Pr\left(\tilde{\mu}_1^{0*} \leq c_A,\ \tilde{\mu}_1^{*} > c_A,\ V_1 > c_E\right), \tag{20}$$

$$p_{act}^{for} = \Pr\left(\tilde{\mu}_1^{0*} > c_A,\ \tilde{\mu}_1^{*} \leq c_A,\ V_1 > c_E\right). \tag{21}$$

Net excess actual entry and gross actual entry distortion rates are then

$$p_{act}^{net} = p_{act}^{exc} - p_{act}^{for}, \qquad p_{act}^{dist} = p_{act}^{exc} + p_{act}^{for}. \tag{22}$$

5.3.2 Characterization

The distortion measures for expected entry, p^{net} and p^{dist}, depend on the model primitives only through the Mincer–Zarnowitz coefficient β_{MZ} and the expected-entry share $r \equiv \Pr(\tilde{\mu}_1^{*} > c_E)$ – see Appendix C for the proofs of claims and propositions in this section. Let $z_0 = \Phi^{-1}(1-r)$ and $\rho = \sqrt{\beta_{\text{MZ}}}$.

[33] The measures capture the effect of the uninformative component of prior heterogeneity. Under overprecision the forecast-level deviation is $(\tilde{\kappa}_0 - \kappa_0)(s_{i0} - \bar{\theta})$, which is perfectly correlated with $\tilde{\mu}_{i1}^{0*}$ rather than orthogonal to it, so the counterfactual would remove this deviation rather than correct the subjective weight $\tilde{\kappa}_0$ back to the Bayesian κ_0.

[34] Excess and forgone applications can be defined analogously, replacing c_E with the application threshold c_A.

Proposition 1. *The expected-entry distortion measures are*

$$p^{net} = \Phi\Big(\tfrac{z_0}{\rho}\Big) - \Phi(z_0), \tag{23}$$

$$p^{dist} = \Phi\Big(\tfrac{z_0}{\rho}\Big) + \Phi(z_0) - 2\,\Phi_2\Big(\tfrac{z_0}{\rho},\, z_0;\, \rho\Big), \tag{24}$$

where $\Phi_2(\cdot,\cdot;\rho)$ *is the standard bivariate normal CDF with correlation* ρ*. When* $r < \frac{1}{2}$*, both measures are strictly decreasing in* β_{MZ}*.*

The actual-entry measures involve two additional inputs: the applicant share, $r_A \equiv \Pr(\tilde{\mu}_1^* > c_A)$, and the scale of business value uncertainty relative to forecast dispersion, $v \equiv \sigma_V/\sigma_{\tilde{\mu}_1^*}$, where $\sigma_V^2 = \text{Var}(V_1) = \sigma_\theta^2 + \sigma_\eta^2$. The applicant share enters here because the heterogeneous prior affects actual entry through the application decision. The standardized thresholds are then functions of the two shares and v,

$$a = \frac{c_A - \bar{\theta}}{\sigma_{\tilde{\mu}_1^*}} = \Phi^{-1}(1 - r_A), \qquad z_V = \frac{c_E - \bar{\theta}}{\sigma_V} = \frac{z_0}{v} = \frac{\Phi^{-1}(1-r)}{v}, \tag{25}$$

so that $\bar{\theta}$, c_A, and c_E need not be separately identified. Let $\overline{\Phi}_2$ and $\overline{\Phi}_3$ denote the bivariate and trivariate standard normal survival functions.

Proposition 2. *The actual-entry distortion measures are*

$$p_{act}^{net} = \overline{\Phi}_2\Big(a,\, z_V;\, \tfrac{\rho^2}{v}\Big) - \overline{\Phi}_2\Big(\tfrac{a}{\rho},\, z_V;\, \tfrac{\rho}{v}\Big), \tag{26}$$

$$p_{act}^{dist} = \overline{\Phi}_2\Big(a,\, z_V;\, \tfrac{\rho^2}{v}\Big) + \overline{\Phi}_2\Big(\tfrac{a}{\rho},\, z_V;\, \tfrac{\rho}{v}\Big) - 2\,\overline{\Phi}_3\Big(\tfrac{a}{\rho},\, a,\, z_V;\, \Sigma\Big), \tag{27}$$

where Σ *is the correlation matrix of* $(\tilde{\mu}_1^{0*}, \tilde{\mu}_1^*, V_1)$*.*

5.3.3 Measurement

We take the values of β_{MZ} for 2017–2021 from the year-by-year two-part model estimates in Table 8. We take the latent pool of potential entrants to be the U.S. resident population aged 24–64 (Pop) – a parsimonious proxy for the working-age population most plausibly at risk of business formation; but the distortion estimates are qualitatively robust to alternative age ranges for the potential-entrant pool (e.g., ages 20-64 or 22-64). The applicant share and the expected-entry share are then given by: $r_A = \Pr(\tilde{\mu}_1^* > c_A) = \text{BA/Pop}$, the ratio of business applications to the pool, and $r = \Pr(\tilde{\mu}_1^* > c_E) = \text{WBA/Pop}$, the ratio of applications with planned wages to the pool

(WBA).[35] These shares pin down the standardized thresholds z_0 and a. The remaining parameter, v, is calibrated to the 4-quarter employer-transition rate of all applicants (7%) over the period 2017-2021. The corresponding rate for WBA (27%) serves as an overidentification check, which the model-implied values (25–28%) pass closely. Appendix D details the calibration and data moments. We report each excess entry and gross entry distortion measures in two units: per capita, i.e., as a share of the population aged 24–64 (the assumed pool of potential applicants), and alternatively, as a share of the applicants per capita.

5.3.4 Results

Table 12: Entry distortions from heterogeneous priors: expected versus actual entry

		A. Expected entry				B. Actual entry			
Year	β_{MZ}	p^{exc} (%)	p^{for} (%)	p^{net}/app (%)	p^{dist}/app (%)	p^{exc}_{act} (%)	p^{for}_{act} (%)	p^{net}_{act}/app (%)	p^{dist}_{act}/app (%)
2017	0.812	0.205	0.032	+9.496	13.047	0.0001	0.0090	−0.493	0.499
2018	0.806	0.215	0.033	+9.181	12.478	0.0003	0.0121	−0.595	0.624
2019	0.789	0.213	0.028	+9.273	12.109	0.0020	0.0188	−0.842	1.046
2020	0.768	0.242	0.028	+8.600	10.872	0.0073	0.0300	−0.911	1.500
2021	0.759	0.296	0.034	+8.474	10.703	0.0089	0.0369	−0.908	1.486

Notes. Panel A reports the expected-entry distortion measures (Definition 1), computed from the closed forms of Proposition 1; Panel B reports the actual-entry distortion measures (Definition 2), computed from Proposition 2. The latent applicant pool is assumed to be the U.S. resident population aged 24–64, so the application share $r_A = \Pr(\tilde{\mu}_1^* > c_A) = \text{BA/Pop}$ and the expected-entry share $r = \text{WBA/Pop}$ are directly observed; the remaining parameter v is calibrated to the 4-quarter employer-transition rate of all applicants (7%). p^{exc}, p^{for} and $p^{exc}_{act}, p^{for}_{act}$ are pool-level probabilities as percent of the population aged 24–64. The "/app" columns divide the net $(p^{exc} - p^{for})$ and gross $(p^{exc} + p^{for})$ distortions by the observed application probability BA/Pop, giving distortion as percent of application rate. Note $p^{net}_{act} < 0$: on net, heterogeneous priors *reduce* actual entry.

Table 12 reveals a sharp sign contrast between the two margins. At the expected-entry margin, heterogeneous priors generate net excess expected entry of 8.5–9.5% of applications and gross distortion of 10.5–13%.[36] At the actual-entry margin, the net effect is negative: heterogeneous priors reduce realized employer entry by 0.49% of applications in 2017 and about 0.91% by 2020–

[35] Note that applicants reporting a planned wage payment are nearly identical to applicants reporting positive expected employment (correlation > 0.99).

[36] Appendix D reports analogous application-margin measures, for which net excess applications are also positive.

2021. This sign reversal reflects selection into application. Applicants induced by ϕ come from the lower tail of the ϕ-free forecast $\tilde{\mu}_1^{0*}$ and rarely become employers, whereas those deterred by ϕ come from the upper tail and are more likely to have entered. Thus, belief noise can increase applications while reducing realized employer entry by changing who selects into application, even though ϕ does not directly affect realized business quality.

The two margins also evolve differently over time. Per applicant, the expected-entry distortion declines modestly as application volume expands, while the actual-entry distortion intensifies: net forgone entry nearly doubles and gross actual-entry distortion roughly triples between 2017 and 2021. Thus, belief noise can raise application activity while worsening selection into subsequent employer entry. These changes are particularly pronounced in 2020–2021, when applications surged and forecast efficiency declined, although the model does not imply that the COVID-19 pandemic itself caused the increase in estimated distortions.

6 Conclusion

This paper links business application-stage forecasts of employer entry and first-year employment to subsequent outcomes in large-scale administrative data, allowing us to quantify expectation errors at both the extensive margin of entry and the intensive margin of initial scale, and to examine how they vary across organizational forms, local conditions, and sectors.

Expectation errors are economically meaningful and highly heterogeneous, even among applicants who both expect and achieve entry. Expectations are biased and inefficient overall, yet informative: expected employment predicts both entry and realized employment, and forecast errors predict near-term outcomes. These patterns point to an important role for heterogeneous priors, learning, and selection at entry, linking canonical firm-dynamics models with theories of entrepreneurial choice and behavioral approaches.

Future work should identify the mechanisms shaping expectations and how different forms of misalignment affect business performance and potential policy interventions. Linking expectations to financing, innovation, hiring, regulation, and different sources of uncertainty could further illuminate early entrepreneurial frictions. The timely, high-frequency BFS data also make expectations potentially useful as early indicators of startup scale and job creation.

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Appendix A Additional tables and figures

Table A-1: Summary statistics for expectation errors by sector

Industry	A. All applications			B. Positive Exp. ($\tilde{\mu}_{it} > 0$)		
	$\tilde{\mu}_{it}$	m_{it}	δ_{it}	$\tilde{\mu}_{it}$	m_{it}	δ_{it}
Accommodation & Food Services	2.142	2.120	0.022	6.994	4.689	2.305
	(0.007)	(0.008)	(0.008)	(0.019)	(0.020)	(0.023)
Administrative & Support & Waste Management	0.609	0.358	0.252	4.352	1.436	2.916
	(0.003)	(0.003)	(0.004)	(0.021)	(0.015)	(0.022)
Arts, Entertainment, & Recreation	0.644	0.442	0.202	5.070	1.958	3.112
	(0.005)	(0.005)	(0.006)	(0.036)	(0.028)	(0.038)
Construction	0.445	0.337	0.108	3.293	1.321	1.972
	(0.002)	(0.002)	(0.002)	(0.011)	(0.009)	(0.012)
Educational Services	0.539	0.352	0.188	4.308	1.591	2.717
	(0.006)	(0.005)	(0.006)	(0.038)	(0.028)	(0.039)
Finance & Insurance	0.286	0.165	0.122	3.903	1.201	2.701
	(0.003)	(0.001)	(0.003)	(0.039)	(0.013)	(0.040)
Health Care & Social Assistance	0.986	0.664	0.322	4.977	1.983	2.995
	(0.005)	(0.004)	(0.005)	(0.021)	(0.015)	(0.021)
Information	0.563	0.248	0.315	4.776	1.055	3.721
	(0.006)	(0.004)	(0.006)	(0.044)	(0.022)	(0.046)
Manufacturing	0.965	0.632	0.333	6.251	2.467	3.785
	(0.009)	(0.007)	(0.010)	(0.052)	(0.036)	(0.052)
Mining, Quarrying, & Oil & Gas Extraction	0.968	0.858	0.111	5.717	2.906	2.811
	(0.028)	(0.030)	(0.034)	(0.149)	(0.128)	(0.161)
Other Services (except Public Administration)	0.399	0.269	0.131	3.442	1.305	2.136
	(0.002)	(0.002)	(0.002)	(0.013)	(0.010)	(0.014)
Professional, Scientific, & Technical Services	0.356	0.273	0.083	3.277	1.358	1.919
	(0.002)	(0.001)	(0.002)	(0.013)	(0.010)	(0.014)
Real Estate & Rental & Leasing	0.174	0.138	0.036	2.531	0.858	1.672
	(0.001)	(0.001)	(0.002)	(0.018)	(0.012)	(0.019)
Retail Trade	0.325	0.218	0.107	3.581	1.466	2.115
	(0.001)	(0.001)	(0.001)	(0.012)	(0.010)	(0.013)
Transportation & Warehousing	0.386	0.257	0.129	3.272	0.975	2.298
	(0.002)	(0.003)	(0.003)	(0.016)	(0.014)	(0.018)
Utilities	0.450	0.279	0.172	5.801	1.814	3.988
	(0.027)	(0.020)	(0.029)	(0.314)	(0.179)	(0.299)
Wholesale Trade	0.431	0.286	0.145	3.673	1.223	2.451
	(0.004)	(0.003)	(0.004)	(0.030)	(0.020)	(0.030)

Notes: Each row corresponds to a 2-digit NAICS industry – excludes Agriculture and Management of Companies & Enterprises. "All" refers to all applications, and "Positive Exp." refers to applications that report positive expected maximum employment. Standard errors are in parentheses. $\delta_{it} = \tilde{\mu}_{it} - m_{it}$.

Table A-2: The definition and sources of covariates

Covariate	**Definition / Construction**	**Data Source**
legal form: sole proprietorship, partnership, corporation, LLC, other	Dummy variables indicating legal form of organization reported on IRS Form SS-4 at application. Sole proprietorship is the omitted category.	BFS microdata (IRS Form SS-4)
prior EIN	Indicator for whether the applicant has previously obtained an EIN.	BFS microdata (IRS Form SS-4)
college share	Share of population aged 25+ with a bachelor's degree or higher in the applicant's census tract, lagged two years.	ACS 5-year estimates
nonwhite share	Share of population in the applicant's census tract identifying as nonwhite, lagged two years.	ACS 5-year estimates
log(median age)	Natural log of median age in the applicant's census tract, lagged two years.	ACS 5-year estimates
log(firm density)	Natural log of number of firms per 1,000 county residents in the applicant's industry–county cell, lagged one year.	LBD
dhs(avg. firm emp.)	Davis–Haltiwanger–Schuh (DHS) transformation of average employment in the applicant's industry–county cell, lagged one year: $DHS(\overline{\text{Emp}})_{j,c,t-1} = \frac{\overline{\text{Emp}}_{j,c,t-1} - \overline{\text{Emp}}}{0.5(\overline{\text{Emp}}_{j,c,t-1} + \overline{\text{Emp}})}$	LBD
top4 emp. share	Employment share of the largest four firms in the applicant's industry–county cell, lagged one year.	LBD
cv(firm emp.)	Coefficient of variation (std. dev. / mean) of firm employment in the applicant's industry–county cell, lagged one year.	LBD
cv(labor prod.)	Coefficient of variation of firm labor productivity (revenue per employee) in the applicant's industry–county cell, lagged one year.	LBD
local gdp growth	County-level GDP growth in the quarter prior to application.	BEA
local covid shock	$(1 - \text{WFHshare}_{j(i)}) \times \frac{\text{NewCases}_{c(i),t}}{\text{Pop}_{c(i),2019}}$, where $\text{WFHshare}_{j(i)}$ is the industry share of jobs teleworkable (Dingel and Neiman, 2020); $\text{NewCases}_{c(i),t}$ is new COVID-19 cases in county c and quarter t; $\text{Pop}_{c(i),2019}$ is pre-pandemic population.	WFHshare: Dingel and Neiman (2020); COVID cases: New York Times; Population: Census; Industry–county mapping from BFS
ltime	Natural log of the number of local procedural days required to legally establish a domestic limited liability company (LLC).	Arizona State University's Doing Business in North America (DBNA) database

Notes: The first column lists the covariates used in the OLS regression in equation (11), the two-part model and the Heckman model in equations (12) and (14); the second column describes the construction of each variable, and the third column reports the corresponding data source.

Table A-3: Two-part model estimation – sample of LLC applications

	A. Probit	B. GLM	
		Gaussian	Gamma
Variable	(1)	(2)	(3)
$\tilde{\mu}_{it}$	0.032***	0.631***	0.494***
	(0.001)	(0.014)	(0.008)
prior EIN	0.378***	0.127	1.090
	(0.088)	(1.045)	(1.170)
college share	0.266***	3.477***	1.495***
	(0.013)	(0.268)	(0.163)
nonwhite share	-0.500***	-0.166	0.379**
	(0.010)	(0.191)	(0.131)
ln(median age)	-0.067***	-0.117	-0.017
	(0.012)	(0.226)	(0.138)
ln(firm density)	0.118***	-0.685***	-0.411***
	(0.007)	(0.137)	(0.090)
dhs(avg. firm emp.)	0.009	0.083	-0.079
	(0.007)	(0.137)	(0.087)
cv(firm emp.)	-0.009***	0.069***	0.047***
	(0.001)	(0.019)	(0.014)
top4 emp. share	0.248***	-2.330***	-1.068**
	(0.032)	(0.597)	(0.379)
cv(labor prod.)	-0.007	0.135	0.039
	(0.006)	(0.123)	(0.071)
local gdp growth	0.006	0.641	-0.917*
	(0.026)	(0.520)	(0.397)
local covid shock	-0.024***	-0.040	-0.070
	(0.004)	(0.065)	(0.038)
const.	-1.725***	-4.902***	-2.409**
	(0.476)	(1.331)	(0.841)
year-quarter FE	Y	Y	Y
industry FE (4-digit NAICS)	Y	Y	Y
state FE	Y	Y	Y
Log Likelihood		-554,600	-466,400
N		1,630,000	1,630,000

Notes: Panel A provides estimates of the Probit model for $\Pr(e_{it} = 1 \mid x_{it}^E)$ in equation (12). Panel B contains estimates of the GLM for $\mathbb{E}[m_{it} \mid e_{it} = 1, x_{it}^M]$ with identity links, assuming Gaussian and Gamma error distributions. Log likelihood is the total log likelihood of Probit and GLM models. Standard errors in parentheses. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. N and log likelihood values are rounded for disclosure avoidance.

Table A-4: Two-part model estimation – sample of LLC applications with positive expectations

	A. Probit	B. GLM	
		Gaussian	Gamma
Variable	(1)	(2)	(3)
$\tilde{\mu}_{it}$	0.007***	0.716***	0.857***
	(0.000)	(0.016)	(0.012)
prior EIN	-0.707***	-3.066	-1.040***
	(0.204)	(1.655)	(0.189)
college share	0.416***	2.440***	0.613***
	(0.027)	(0.408)	(0.183)
nonwhite share	-0.681***	-0.479	0.084
	(0.021)	(0.306)	(0.145)
ln(median age)	-0.033	0.064	0.266
	(0.023)	(0.345)	(0.156)
ln(firm density)	0.116***	-0.800***	-0.205
	(0.015)	(0.222)	(0.113)
dhs(avg. firm emp.)	-0.007	0.088	0.014
	(0.015)	(0.208)	(0.099)
cv(firm emp.)	-0.014***	0.018	0.011
	(0.002)	(0.031)	(0.017)
top4 emp. share	0.398***	-2.011*	-0.971*
	(0.066)	(0.896)	(0.419)
cv(labor prod.)	-0.029**	0.190	0.060
	(0.011)	(0.175)	(0.078)
local gdp growth	-0.043	0.465	0.197
	(0.048)	(0.704)	(0.386)
local covid shock	-0.015*	0.004	0.013
	(0.007)	(0.103)	(0.048)
const.	0.194	-2.407	-0.539
	(0.188)	(2.545)	(1.332)
year-quarter FE	Y	Y	Y
industry FE (4-digit NAICS)	Y	Y	Y
state FE	Y	Y	Y
Log Likelihood		-193,800	-161,000
N		178,000	178,000

Notes: Panel A provides estimates of the Probit model for $\Pr(e_{it} = 1 \mid x_{it}^E)$ in equation (12). Panel B contains estimates of the GLM for $\mathbb{E}[m_{it} \mid e_{it} = 1, x_{it}^M]$ with identity links, assuming Gaussian and Gamma error distributions. Log likelihood is the total log likelihood of Probit and GLM models. Standard errors in parentheses. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. N and log likelihood values are rounded for disclosure avoidance.

Appendix B Robustness to measurement frequency

To assess the sensitivity of the estimates of the average expectation error to perceived employment measurement frequency by the potential entrant, we estimate theoretical upper bounds for the expected value of the maximum of τ random variables where $\tau = 12, 24$, corresponding to monthly and approximately bi-weekly measurement frequencies for realized employment.

First, we use a two-part model to estimate the probability of entry, $\Pr(e_{it} = 1)$ and the average realized employment (conditional on entry), $\hat{\mu}_{it}$, over the 4 quarter horizon as a function of observables. We also use an OLS regression to estimate the standard deviation of realized quarterly employment across quarters conditional on entry, again using application characteristics and fixed effects as predictors. We then obtain the application-level predicted average quarterly employment, $\hat{\mu}_{it}$, and the predicted standard deviation of quarterly employment, $\hat{\sigma}_{it}$.

We also estimate the average pairwise correlation between two consecutive quarterly employment levels conditional on entry, which yields $\hat{\rho}_e = 0.87$ – indicating that the correlation between two consecutive quarterly employment levels is quite high. As an alternative, we use a higher (assumed) value $\hat{\rho}_e = 0.95$ to capture the likely higher average correlation between two consecutive monthly or bi-weekly employment levels.

The estimated theoretical upper bound for the expected value of the maximum of τ Gaussian (not necessarily independent) random variables is then given by

$$\hat{U}(\mathbb{E}[m_{it}|e_{it} = 1]) = \hat{\mu}_{it} + \hat{\sigma}_{it} \times \sqrt{2(1 - \hat{\rho}_e)\tau}. \tag{B-1}$$

A similar estimated upper bound can also be calculated for the case of Gamma random variables as

$$\hat{U}(\mathbb{E}[m_{it}|e_{it} = 1]) = \frac{2(\hat{\sigma}_{it}/\hat{\mu}_{it})(1 - \hat{\rho}_e)ln(\tau)}{1 - \tau^{-\hat{\mu}^2_{it}/\hat{\sigma}_{it}}}. \tag{B-2}$$

We then construct, for each application, an estimate of the theoretical upper bound of the expected value of realized maximum employment (m_{it}) in the 4 quarters following application

$$\begin{aligned} \hat{U}(\mathbb{E}[m_{it}]) &= \widehat{\Pr}(e_{it} = 1) \times \hat{U}(\mathbb{E}[m_{it}|e_{it} = 1]) + (1 - \widehat{\Pr}(e_{it} = 1)) \times 0 \\ &= \widehat{\Pr}(e_{it} = 1) \times \hat{U}(\mathbb{E}[m_{it}|e_{it} = 1]). \end{aligned} \tag{B-3}$$

Using (B-3), we construct estimates of the average expectation error $\hat{\bar{\delta}}_\tau$ for $\tau = 12, 24$. For comparison, we also calculate the average expectation errors based on *i.i.d.* assumption ($\hat{\rho}_e = 0$) for quarterly employments, $\hat{\bar{\delta}}^{iid}_\tau$ for each case $\tau = 12, 24$. We also report the average predicted expectation error based on the estimated maximum employment using the two-part model, $\hat{\bar{\delta}}^p$, for comparison with the average (raw) expectation error.

The results in Tables B-1 and B-2 indicate that estimated average expectation error using the theoretical upper bounds are significantly greater than zero statistically for both cases, $\rho_e = 0.87, 0.95$ and only becomes negative when the extreme assumption of *i.i.d* quarterly employments is imposed. These results support the finding of overestimation with quarterly measurement, given the fact that these estimated upper bounds are conservative (i.e., larger than the actual theoretical expected value of the maximum employment).

Table B-1: Expectation errors based on theoretical upper bounds for maximum employment — All applications

	A. Gaussian		B. Gamma	
Statistic	$\hat{\rho}_e = 0.87$	$\hat{\rho}_e = 0.95$	$\hat{\rho}_e = 0.87$	$\hat{\rho}_e = 0.95$
$\hat{\bar{\delta}}$	0.144	0.144	0.144	0.144
	(0.0009)	(0.0009)	(0.0009)	(0.0009)
$\hat{\bar{\delta}}_p$	0.144	0.144	0.144	0.144
	(0.0006)	(0.0006)	(0.0005)	(0.0005)
$\hat{\bar{\delta}}_{12}$	0.045	0.119	0.025	0.160
	(0.0006)	(0.0007)	(0.0006)	(0.0007)
$\hat{\bar{\delta}}_{24}$	0.019	0.103	0.022	0.142
	(0.0006)	(0.0006)	(0.0006)	(0.0007)
$\hat{\bar{\delta}}_{12}^{iid}$	-0.305	-0.305	-1.439	-1.439
	(0.0006)	(0.0006)	(0.0007)	(0.0007)
$\hat{\bar{\delta}}_{24}^{iid}$	-0.376	-0.376	-1.799	-1.799
	(0.0006)	(0.0006)	(0.0008)	(0.0008)

Notes: The table shows mean expectation errors and associated standard errors (in parentheses). Results are reported for two values of the quarterly employment correlation, $\hat{\rho}_e = 0.87$ and $\hat{\rho}_e = 0.95$, under both Gaussian and Gamma forecast error assumptions. The value $\hat{\rho}_e = 0.87$ corresponds to the average observed correlation between consecutive quarterly employment levels within a firm, while $\hat{\rho}_e = 0.95$ reflects an assumed correlation for monthly employment.

Table B-2: Expectation errors based on theoretical upper bounds for maximum employment — Applications with positive exp. ($\tilde{\mu}_{it} > 0$)

	A. Gaussian		B. Gamma	
Statistic	$\hat{\rho}_e = 0.87$	$\hat{\rho}_e = 0.95$	$\hat{\rho}_e = 0.87$	$\hat{\rho}_e = 0.95$
$\hat{\bar{\delta}}$	2.436	2.436	2.436	2.436
	(0.006)	(0.006)	(0.006)	(0.006)
$\hat{\bar{\delta}}_p$	2.437	2.437	2.433	2.433
	(0.004)	(0.004)	(0.004)	(0.004)
$\hat{\bar{\delta}}_{12}$	2.136	2.402	2.153	2.583
	(0.004)	(0.004)	(0.004)	(0.004)
$\hat{\bar{\delta}}_{24}$	2.044	2.345	2.047	2.542
	(0.004)	(0.004)	(0.004)	(0.004)
$\hat{\bar{\delta}}_{12}^{iid}$	0.891	0.891	-2.521	-2.521
	(0.004)	(0.004)	(0.004)	(0.004)
$\hat{\bar{\delta}}_{24}^{iid}$	0.636	0.636	-3.341	-3.341
	(0.004)	(0.004)	(0.005)	(0.005)

Notes: The table shows mean expectation errors and associated standard errors (in parentheses). Results are reported for two values of the quarterly employment correlation, $\hat{\rho}_e = 0.87$ and $\hat{\rho}_e = 0.95$, under both Gaussian and Gamma forecast error assumptions. The value $\hat{\rho}_e = 0.87$ corresponds to the average observed correlation between consecutive quarterly employment levels within a firm, while $\hat{\rho}_e = 0.95$ reflects an assumed correlation for monthly employment.

Appendix C Proofs

C.1 Implication (i): Mincer–Zarnowitz slope $\in (0, 1)$

Proof. The Mincer-Zarnowitz slope can be expressed as

$$\beta_{\mathrm{MZ}} = \frac{\mathrm{Cov}(m_1^*, \tilde{\mu}_1^*)}{\mathrm{Var}(\tilde{\mu}_1^*)}.$$

Since $\tilde{\mu}_1^* = (1-\kappa_0)\bar{\theta} + \kappa_0(\theta+\varepsilon_0) + (1-\kappa_0)\phi$ and $\theta, \varepsilon_0, \varepsilon_1, \phi$ are mutually independent, with $\varepsilon_0, \varepsilon_1, \phi$ centered,

$$\mathrm{Cov}(m_1^*, \tilde{\mu}_1^*) = \mathrm{Cov}(\theta + \varepsilon_1,\ \kappa_0(\theta+\varepsilon_0) + (1-\kappa_0)\phi) = \kappa_0 \,\mathrm{Var}(\theta) = \kappa_0 \sigma_\theta^2.$$

For the variance of $\tilde{\mu}_1^*$,

$$\mathrm{Var}(\tilde{\mu}_1^*) = \mathrm{Var}\big(\kappa_0(\theta+\varepsilon_0)\big) + \mathrm{Var}\big((1-\kappa_0)\phi\big) = \kappa_0^2(\sigma_\theta^2 + \sigma_0^2) + (1-\kappa_0)^2\sigma_\phi^2 = \kappa_0\sigma_\theta^2 + (1-\kappa_0)^2\sigma_\phi^2,$$

where the last equality uses $\kappa_0 = \frac{\sigma_\theta^2}{\sigma_\theta^2+\sigma_0^2}$. Therefore

$$\beta_{\mathrm{MZ}} = \frac{\kappa_0\sigma_\theta^2}{\kappa_0\sigma_\theta^2 + (1-\kappa_0)^2\sigma_\phi^2}.$$

If $\sigma_0 = +\infty$, then $\kappa_0 = 0$ and $\beta_{\mathrm{MZ}} = 0$. If $\sigma_\phi^2 = 0$, the denominator equals the numerator and $\beta_{\mathrm{MZ}} = 1$. If $\sigma_0 < +\infty$ and $\sigma_\phi^2 > 0$, then the denominator strictly exceeds the numerator, hence $0 < \beta_{\mathrm{MZ}} < 1$. Absent both an informative signal and prior heterogeneity, $\tilde{\mu}_1^*$ is degenerate and the slope is undefined. Note that σ_ε^2 drops out because it affects m_1^* only through an innovation independent of $\tilde{\mu}_1^*$, hence it does not enter the covariance.

Now, we prove β_{MZ} does not change when conditional on the set of applicants $A = \{\tilde{\mu}_1^* > c_A\}$, or the set of applicants with positive expected employment $B = \{\tilde{\mu}_1^* > c_E\}$. Because $(m_1^*, \tilde{\mu}_1^*)$ is jointly normal, $\mathbb{E}[m_1^* \mid \tilde{\mu}_1^*] = \alpha_{\mathrm{MZ}} + \beta_{\mathrm{MZ}}\,\tilde{\mu}_1^*$. Define $\nu = m_1^* - (\alpha_{\mathrm{MZ}} + \beta_{\mathrm{MZ}}\,\tilde{\mu}_1^*)$. Then $\mathbb{E}[\nu \mid \tilde{\mu}_1^*] = 0$. Because A and B are selection events measurable with respect to $\tilde{\mu}_1^*$, we have $\mathbb{E}[\nu \mid A] = \mathbb{E}[\nu \mid B] = 0$ and $\mathrm{Cov}(\nu, \tilde{\mu}_1^* \mid A) = \mathrm{Cov}(\nu, \tilde{\mu}_1^* \mid B) = 0$. Hence,

$$\mathrm{Cov}(m_1^*, \tilde{\mu}_1^* \mid A) = \mathrm{Cov}\big(\alpha_{\mathrm{MZ}} + \beta_{\mathrm{MZ}}\tilde{\mu}_1^* + \nu,\ \tilde{\mu}_1^* \mid A\big) = \beta_{\mathrm{MZ}}\,\mathrm{Var}(\tilde{\mu}_1^* \mid A),$$

$$\mathrm{Cov}(m_1^*, \tilde{\mu}_1^* \mid B) = \mathrm{Cov}\big(\alpha_{\mathrm{MZ}} + \beta_{\mathrm{MZ}}\tilde{\mu}_1^* + \nu,\ \tilde{\mu}_1^* \mid B\big) = \beta_{\mathrm{MZ}}\,\mathrm{Var}(\tilde{\mu}_1^* \mid B).$$

So the Mincer–Zarnowitz slope among applicants and among applicants with positive expected employment equals the population slope:

$$\beta_{\mathrm{MZ}}^{(A)} = \frac{\mathrm{Cov}(m_1^*, \tilde{\mu}_1^* \mid A)}{\mathrm{Var}(\tilde{\mu}_1^* \mid A)} = \beta_{\mathrm{MZ}}, \qquad \beta_{\mathrm{MZ}}^{(B)} = \frac{\mathrm{Cov}(m_1^*, \tilde{\mu}_1^* \mid B)}{\mathrm{Var}(\tilde{\mu}_1^* \mid B)} = \beta_{\mathrm{MZ}}.$$

C.2 Implication (ii): Predictive content of expectation errors

Proof. Let $X_i = (\theta_i, \eta_{i1}, \varepsilon_{i1}, \{\eta_{ik}, \varepsilon_{ik}\}_{k\geq 2})$ collect latent size and all entry and employment shocks. In the model of Section 5, realized entry and employment, $e_{ik} = \mathbf{1}\{\theta_i + \eta_{ik} > c_E\}$ and $m_{ik} = e_{ik}(\theta_i + \varepsilon_{ik})$ for $k \geq 1$, are functions of X_i; the prior shift ϕ_i is independent of X_i; and the application decision, the expected-entry decision, and the reported forecast $\tilde{\mu}_{i1} = \tilde{e}_{i1}\tilde{\mu}_{i1}^*$ are functions of $\tilde{\mu}_{i1}^*$. The four estimation samples are

$$\mathcal{S}_1 = \{\tilde{a}_{i1} = 1\}, \qquad \mathcal{S}_2 = \mathcal{S}_1 \cap \{\tilde{\mu}_{i1} > 0\}, \qquad \mathcal{S}_3 = \mathcal{S}_1 \cap \{m_{i1} > 0\}, \qquad \mathcal{S}_4 = \mathcal{S}_2 \cap \{m_{i1} > 0\}.$$

For a sample $\mathcal{S}$, let $\beta_2^{\mathcal{S}}$ be the population OLS coefficient on δ_{i1} in the regression of m_{ik} on $(1, m_{i1}, \delta_{i1})$ within $\mathcal{S}$, the probability limit of the estimate in (15).

Lemma 1. *Let $k \geq 2$ and $\mathcal{S} \in \{\mathcal{S}_1, \ldots, \mathcal{S}_4\}$, with $\mathrm{Var}(m_{i1} \mid \mathcal{S}) > 0$ and $\mathrm{Var}(\tilde{\mu}_{i1} \mid \mathcal{S}) > 0$. If $\kappa_0 = 0$, then $\beta_2^{\mathcal{S}} = 0$.*

Proof. If $\kappa_0 = 0$, then $\tilde{\mu}_{i1}^* = \bar{\theta} + \phi_i$, so the reported forecast $\tilde{\mu}_{i1}$ and the application and expected-entry decisions are functions of ϕ_i alone, whereas m_{i1} and m_{ik} are functions of X_i alone. Each sample restricts either the forecast or first-year employment, so it can be written as $\mathcal{S} = \{\phi_i \in F\} \cap \{X_i \in G\}$ for measurable sets F and G. Because ϕ_i and X_i are independent, for any measurable F' and G',

$$\begin{aligned}\Pr(\phi_i \in F',\, X_i \in G' \mid \mathcal{S}) &= \frac{\Pr(\phi_i \in F' \cap F)\, \Pr(X_i \in G' \cap G)}{\Pr(\phi_i \in F)\, \Pr(X_i \in G)} \\ &= \Pr(\phi_i \in F' \mid \phi_i \in F)\, \Pr(X_i \in G' \mid X_i \in G),\end{aligned}$$

so ϕ_i and X_i remain independent within $\mathcal{S}$, and hence $\tilde{\mu}_{i1}$ is independent of (m_{i1}, m_{ik}) within $\mathcal{S}$. Since $\delta_{i1} = \tilde{\mu}_{i1} - m_{i1}$ and m_{i1} is a regressor, $\beta_2^{\mathcal{S}}$ equals the coefficient on $\tilde{\mu}_{i1}$ in the regression of m_{ik} on $(1, m_{i1}, \tilde{\mu}_{i1})$ within $\mathcal{S}$. The two regressors are uncorrelated within $\mathcal{S}$, so their covariance matrix is diagonal and the normal equations give

$$\beta_2^{\mathcal{S}} = \frac{\mathrm{Cov}(m_{ik}, \tilde{\mu}_{i1} \mid \mathcal{S})}{\mathrm{Var}(\tilde{\mu}_{i1} \mid \mathcal{S})} = 0,$$

where the last equality follows from the independence of $\tilde{\mu}_{i1}$ and m_{ik} within $\mathcal{S}$. □

By contraposition, $\beta_2^{\mathcal{S}} \neq 0$ in any sample implies $\kappa_0 > 0$: the reported forecast contains information about latent size beyond first-year employment. The proof uses only that, absent an informative signal, the forecast depends on ϕ_i alone, that realized outcomes depend on X_i alone, and that ϕ_i is independent of X_i. It requires no normality, no functional form for e_{ik} or m_{ik}, and no independence of shocks over time, so it applies to any outcome determined by X_i, including the employer-status indicator e_{ik}. The same argument holds when the regression includes controls z_i, provided ϕ_i is independent of (X_i, z_i).

C.3 Implication (iii): Overestimation among applicants and underestimation among entrants

Proof. *Part 1: latent error among applicants.* Since $m_1^* = \theta + \varepsilon_1$ with ε_1 independent of $\tilde{\mu}_1^*$ and centered, $\mathbb{E}[\delta_1^* \mid \tilde{\mu}_1^*] = \tilde{\mu}_1^* - \mathbb{E}[\theta \mid \tilde{\mu}_1^*]$. Since $\theta, \varepsilon_0, \phi$ are mutually independent normal random variables, $(\theta, \tilde{\mu}_1^*)$ is jointly Gaussian and

$$\mathbb{E}[\theta \mid \tilde{\mu}_1^*] = \bar{\theta} + \frac{\mathrm{Cov}(\theta, \tilde{\mu}_1^*)}{\mathrm{Var}(\tilde{\mu}_1^*)}\,(\tilde{\mu}_1^* - \bar{\theta}).$$

The required moments are $\mathbb{E}[\tilde{\mu}_1^*] = \bar{\theta}$, $\mathrm{Cov}(\theta, \tilde{\mu}_1^*) = \mathrm{Cov}(\theta, \kappa_0\theta) = \kappa_0\sigma_\theta^2$ by independence and centering of ε_0, ϕ, and

$$\sigma_{\tilde{\mu}_1^*}^2 = \mathrm{Var}(\tilde{\mu}_1^*) = \kappa_0^2(\sigma_\theta^2 + \sigma_0^2) + (1-\kappa_0)^2\sigma_\phi^2 = \kappa_0\sigma_\theta^2 + (1-\kappa_0)^2\sigma_\phi^2,$$

the last equality by $\kappa_0 = \sigma_\theta^2/(\sigma_\theta^2 + \sigma_0^2)$. Hence

$$\mathbb{E}[\delta_1^* \mid \tilde{\mu}_1^*] = \left(1 - \frac{\kappa_0\sigma_\theta^2}{\sigma_{\tilde{\mu}_1^*}^2}\right)(\tilde{\mu}_1^* - \bar{\theta}) = \frac{(1-\kappa_0)^2\sigma_\phi^2}{\sigma_{\tilde{\mu}_1^*}^2}\,(\tilde{\mu}_1^* - \bar{\theta}).$$

Since $\tilde{\mu}_1^* \sim \mathcal{N}(\bar{\theta}, \sigma_{\tilde{\mu}_1^*}^2)$, the truncated-normal mean $\mathbb{E}[\tilde{\mu}_1^* - \bar{\theta} \mid \tilde{\mu}_1^* > c_A] = \sigma_{\tilde{\mu}_1^*}\lambda\big((c_A - \bar{\theta})/\sigma_{\tilde{\mu}_1^*}\big)$ gives

$$\mathbb{E}[\delta_1^* \mid \tilde{\mu}_1^* > c_A] = \frac{(1-\kappa_0)^2\sigma_\phi^2}{\sigma_{\tilde{\mu}_1^*}}\,\lambda\!\left(\frac{c_A - \bar{\theta}}{\sigma_{\tilde{\mu}_1^*}}\right) > 0,$$

as $(1-\kappa_0)^2\sigma_\phi^2 > 0$ for $\sigma_0^2, \sigma_\phi^2 \in (0,\infty)$ and $\lambda(\cdot) > 0$.

Part 2: latent error among entrants. Actual entry is $\{V_1 > c_E\}$ with $V_1 = \theta + \eta_1$, so $(\tilde{\mu}_1^*, m_1^*, V_1)$ is jointly Gaussian with common mean $\bar{\theta}$, $\mathrm{Cov}(\tilde{\mu}_1^*, V_1) = \kappa_0\sigma_\theta^2$, $\mathrm{Cov}(m_1^*, V_1) = \sigma_\theta^2 + \rho_{\eta\varepsilon}\sigma_\eta\sigma_\varepsilon$ and $\mathrm{Var}(V_1) = \sigma_V^2 = \sigma_\theta^2 + \sigma_\eta^2$. Applying the truncated-normal mean formula to $(\tilde{\mu}_1^*, V_1)$ and to (m_1^*, V_1), with $z_V = (c_E - \bar{\theta})/\sigma_V$,

$$\mathbb{E}[\delta_1^* \mid V_1 > c_E] = \frac{\lambda(z_V)}{\sigma_V}\Big[\kappa_0\sigma_\theta^2 - \sigma_\theta^2 - \rho_{\eta\varepsilon}\sigma_\eta\sigma_\varepsilon\Big] = -\frac{(1-\kappa_0)\sigma_\theta^2 + \rho_{\eta\varepsilon}\sigma_\eta\sigma_\varepsilon}{\sigma_V}\,\lambda(z_V),$$

strictly negative *if and only if*

$$(1-\kappa_0)\sigma_\theta^2 + \rho_{\eta\varepsilon}\sigma_\eta\sigma_\varepsilon > 0, \tag{C1}$$

which always holds when $\rho_{\eta\varepsilon} \geq 0$ and otherwise requires that the negative correlation between the entry and employment shocks not overturn the θ-driven truncation.

Part 3: latent error among applicants who expect and achieve entry. Let $\mathcal{S} = \{\tilde{\mu}_1^* > c_E,\ V_1 > c_E\}$, which lies within the set of applicants since $c_E > c_A$. Standardize $Z_1 = (\tilde{\mu}_1^* - \bar{\theta})/\sigma_{\tilde{\mu}_1^*}$ and $Z_2 = (V_1 - \bar{\theta})/\sigma_V$, with $\rho_{\tilde{\mu}V} = \mathrm{Corr}(\tilde{\mu}_1^*, V_1) = \kappa_0\sigma_\theta^2/(\sigma_{\tilde{\mu}_1^*}\sigma_V)$ and $z_0 = (c_E - \bar{\theta})/\sigma_{\tilde{\mu}_1^*}$. For any W jointly normal with (Z_1, Z_2), the means of the truncated bivariate normal (Tallis, 1961) give

$$\mathbb{E}[W \mid \mathcal{S}] = \mathbb{E}[W] + \frac{\mathrm{Cov}(W, Z_1)A_1 + \mathrm{Cov}(W, Z_2)A_2}{\overline{\Phi}_2(z_0, z_V; \rho_{\tilde{\mu}V})}, \quad A_1 = \varphi(z_0)\,\overline{\Phi}\Big(\tfrac{z_V - \rho_{\tilde{\mu}V}z_0}{\sqrt{1-\rho_{\tilde{\mu}V}^2}}\Big),\ A_2 = \varphi(z_V)\,\overline{\Phi}\Big(\tfrac{z_0 - \rho_{\tilde{\mu}V}z_V}{\sqrt{1-\rho_{\tilde{\mu}V}^2}}\Big),$$

where $\overline{\Phi}$ and $\overline{\Phi}_2$ are the univariate and bivariate standard normal survival functions. With $W = \delta_1^*$, $\mathbb{E}[\delta_1^*] = 0$, $\mathrm{Cov}(\delta_1^*, Z_1) = (1-\kappa_0)^2\sigma_\phi^2/\sigma_{\tilde{\mu}_1^*}$, and $\mathrm{Cov}(\delta_1^*, Z_2) = -[(1-\kappa_0)\sigma_\theta^2 + \rho_{\eta\varepsilon}\sigma_\eta\sigma_\varepsilon]/\sigma_V$, so

$$\mathbb{E}[\delta_1^* \mid \mathcal{S}] = \frac{1}{\overline{\Phi}_2(z_0, z_V; \rho_{\tilde{\mu}V})}\left[\frac{(1-\kappa_0)^2\sigma_\phi^2}{\sigma_{\tilde{\mu}_1^*}}A_1 - \frac{(1-\kappa_0)\sigma_\theta^2 + \rho_{\eta\varepsilon}\sigma_\eta\sigma_\varepsilon}{\sigma_V}A_2\right].$$

The first term reflects selection on the forecast, which raises the uninformative component ϕ; the second reflects selection on entry, as in Part 2. Letting $z_V \to -\infty$ or $z_0 \to -\infty$ recovers Parts 1 and 2. Hence $\mathbb{E}[\delta_1^* \mid \mathcal{S}] < 0$ if and only if

$$(1-\kappa_0)\sigma_\theta^2 + \rho_{\eta\varepsilon}\sigma_\eta\sigma_\varepsilon > \frac{\sigma_V}{\sigma_{\tilde{\mu}_1^*}}(1-\kappa_0)^2\sigma_\phi^2\,\frac{A_1}{A_2}. \tag{C2}$$

Since the right-hand side is positive, (C2) implies (C1): entry selection must outweigh selection on the uninformative prior.

C.4 Proof of Proposition 1

Proof. In the Gaussian signal structure, the latent forecast and its ϕ-free counterpart are

$$\tilde{\mu}^* = (1-\kappa_0)(\bar{\theta} + \phi) + \kappa_0 s_0, \qquad \tilde{\mu}^{0*} = (1-\kappa_0)\bar{\theta} + \kappa_0 s_0,$$

so that $\tilde{\mu}^* = \tilde{\mu}^{0*} + (1-\kappa_0)\phi$. The latent outcome is $m^* = \theta + \varepsilon$ (with ε independent of all else), and the Mincer–Zarnowitz coefficient is $\beta_{\mathrm{MZ}} = \mathrm{Cov}(m^*, \tilde{\mu}^*)/\mathrm{Var}(\tilde{\mu}^*)$. For the threshold c_E, define

$$r = \Pr(\tilde{\mu}^* > c_E), \qquad r^0 = \Pr(\tilde{\mu}^{0*} > c_E),$$

$$p^{exc} = \Pr(\tilde{\mu}^{0*} \leq c_E < \tilde{\mu}^*), \qquad p^{for} = \Pr(\tilde{\mu}^* \leq c_E < \tilde{\mu}^{0*}),$$

$$p^{net} = p^{exc} - p^{for} = r - r^0, \qquad p^{dist} = p^{exc} + p^{for}.$$

Step 1: the joint law of $(\tilde{\mu}^{0*}, \tilde{\mu}^*)$. Since $\mathbb{E}[\phi] = 0$ and $\mathbb{E}[s_0] = \bar{\theta}$, both beliefs have mean $\bar{\theta}$. As $\tilde{\mu}^{0*}$ is a function of (θ, ε_0) and ϕ is independent of them,

$$\mathrm{Var}(\tilde{\mu}^{0*}) = \kappa_0^2(\sigma_\theta^2 + \sigma_0^2) = \kappa_0\sigma_\theta^2, \qquad (1-\kappa_0)\phi \perp \tilde{\mu}^{0*}.$$

Hence $\mathrm{Cov}(\tilde{\mu}^{0*}, \tilde{\mu}^*) = \mathrm{Var}(\tilde{\mu}^{0*}) = \kappa_0\sigma_\theta^2$ and $\mathrm{Var}(\tilde{\mu}^*) = \kappa_0\sigma_\theta^2 + (1-\kappa_0)^2\sigma_\phi^2$. Moreover $\mathrm{Cov}(m^*, \tilde{\mu}^*) = \mathrm{Cov}(\theta, \tilde{\mu}^*) = \kappa_0\sigma_\theta^2$, so

$$\beta_{\mathrm{MZ}} = \frac{\mathrm{Cov}(m^*, \tilde{\mu}^*)}{\mathrm{Var}(\tilde{\mu}^*)} = \frac{\kappa_0\sigma_\theta^2}{\mathrm{Var}(\tilde{\mu}^*)} = \frac{\mathrm{Var}(\tilde{\mu}^{0*})}{\mathrm{Var}(\tilde{\mu}^*)}.$$

Therefore, the correlation between the ϕ-free belief and the latent forecast is exactly the square root of the MZ coefficient:

$$\mathrm{Corr}(\tilde{\mu}^{0*}, \tilde{\mu}^*) = \frac{\mathrm{Cov}(\tilde{\mu}^{0*}, \tilde{\mu}^*)}{\sqrt{\mathrm{Var}(\tilde{\mu}^{0*})\,\mathrm{Var}(\tilde{\mu}^*)}} = \sqrt{\frac{\mathrm{Var}(\tilde{\mu}^{0*})}{\mathrm{Var}(\tilde{\mu}^*)}} = \sqrt{\beta_{\mathrm{MZ}}} \;\equiv\; \rho \in (0,1).$$

Step 2: (β_{MZ}, r) as sufficient statistics. $(\tilde{\mu}^{0*}, \tilde{\mu}^*)$ is jointly normal, being a linear image of $(\theta, \varepsilon_0, \phi)$. Standardize:

$$\zeta = \frac{\tilde{\mu}^* - \bar{\theta}}{\sigma_{\tilde{\mu}^*}} \sim \mathcal{N}(0,1), \qquad \zeta^0 = \frac{\tilde{\mu}^{0*} - \bar{\theta}}{\sqrt{\mathrm{Var}(\tilde{\mu}^{0*})}} = \frac{\tilde{\mu}^{0*} - \bar{\theta}}{\rho\,\sigma_{\tilde{\mu}^*}} \sim \mathcal{N}(0,1), \qquad \mathrm{Corr}(\zeta^0, \zeta) = \rho.$$

Let $z_0 \equiv (c_E - \bar{\theta})/\sigma_{\tilde{\mu}^*}$ be the standardized threshold, so that $r = \Pr(\zeta > z_0) = 1 - \Phi(z_0)$, i.e. $z_0 = \Phi^{-1}(1-r)$. Because $\mathrm{Var}(\tilde{\mu}^{0*}) = \rho^2\sigma_{\tilde{\mu}^*}^2$, the event $\{\tilde{\mu}^{0*} > c_E\}$ is $\{\zeta^0 > z_0/\rho\}$. Writing Φ for the standard normal CDF and $\Phi_2(\cdot,\cdot;\rho)$ for the standard bivariate normal CDF with correlation ρ,

$$r = 1 - \Phi(z_0), \qquad r^0 = 1 - \Phi(z_0/\rho),$$

$$p^{exc} = \Phi(z_0/\rho) - \Phi_2(z_0/\rho, z_0; \rho), \qquad p^{for} = \Phi(z_0) - \Phi_2(z_0/\rho, z_0; \rho),$$

and therefore

$$p^{net} = \Phi(z_0/\rho) - \Phi(z_0), \qquad p^{dist} = \Phi(z_0/\rho) + \Phi(z_0) - 2\,\Phi_2(z_0/\rho, z_0; \rho).$$

Since $z_0 = \Phi^{-1}(1-r)$ depends only on r and $\rho = \sqrt{\beta_{\mathrm{MZ}}}$, both p^{net} and p^{dist} are functions of (β_{MZ}, r) alone; no other primitive enters. This proves the first claim.

Step 3: monotonicity in β_{MZ}. Fix r (hence z_0) and view the measures as functions of ρ. As $d\rho/d\beta_{\mathrm{MZ}} = 1/(2\sqrt{\beta_{\mathrm{MZ}}}) > 0$, the sign of any derivative in β_{MZ} equals its sign in ρ. Write $y \equiv z_0/\rho$, so $dy/d\rho = -z_0/\rho^2$. We use two standard facts (φ is the standard normal pdf, φ_2 the standard bivariate normal density):

$$\text{(i)}\ \frac{\partial\Phi_2(h,k;\rho)}{\partial h} = \varphi(h)\,\Phi\Big(\tfrac{k-\rho h}{\sqrt{1-\rho^2}}\Big), \qquad \text{(ii)}\ \frac{\partial\Phi_2(h,k;\rho)}{\partial\rho} = \varphi_2(h,k;\rho).$$

At $(h,k) = (z_0/\rho, z_0)$ the cross term vanishes, since $k - \rho h = z_0 - \rho(z_0/\rho) = 0$ gives $\Phi(0) = \frac{1}{2}$; and because $y^2 - 2\rho y z_0 + z_0^2 = z_0^2(1-\rho^2)/\rho^2$,

$$\varphi_2(y, z_0; \rho) = \frac{1}{2\pi\sqrt{1-\rho^2}}\, e^{-z_0^2/(2\rho^2)} = \frac{\varphi(y)}{\sqrt{2\pi(1-\rho^2)}}.$$

Net excess entry. From $p^{net} = \Phi(y) - \Phi(z_0)$,

$$\frac{d\,p^{net}}{d\rho} = \varphi(y)\,\frac{dy}{d\rho} = -\frac{z_0}{\rho^2}\,\varphi(y).$$

If $r < \frac{1}{2}$ then $z_0 = \Phi^{-1}(1-r) > 0$; with $\varphi(y) > 0$ this is strictly negative. *Gross distortion.* Differentiating $\Phi_2(y, z_0; \rho)$ through both its first argument and ρ,

$$\frac{d}{d\rho}\Phi_2(y, z_0; \rho) = \varphi(y)\,\Phi(0)\Big(-\tfrac{z_0}{\rho^2}\Big) + \frac{\varphi(y)}{\sqrt{2\pi(1-\rho^2)}} = -\tfrac{z_0}{2\rho^2}\,\varphi(y) + \frac{\varphi(y)}{\sqrt{2\pi(1-\rho^2)}}.$$

Hence, from $p^{dist} = \Phi(y) + \Phi(z_0) - 2\Phi_2(y, z_0; \rho)$,

$$\frac{d\,p^{dist}}{d\rho} = \varphi(y)\Big(-\tfrac{z_0}{\rho^2}\Big) - 2\left[-\tfrac{z_0}{2\rho^2}\varphi(y) + \frac{\varphi(y)}{\sqrt{2\pi(1-\rho^2)}}\right] = -\frac{2\,\varphi(y)}{\sqrt{2\pi(1-\rho^2)}} < 0,$$

where the two z_0/ρ^2 terms cancel exactly. Since the sign in β_{MZ} matches that in ρ, p^{dist} is strictly decreasing in β_{MZ}, and p^{net} is strictly decreasing in β_{MZ} whenever $r < \frac{1}{2}$.

C.5 Proof of Proposition 2

Proof. Step 1 of the proof of Proposition 1 established that $(\tilde{\mu}_1^{0*}, \tilde{\mu}_1^{*})$ is jointly normal with common mean $\bar{\theta}$, $\mathrm{Var}(\tilde{\mu}_1^{0*}) = \kappa_0\sigma_\theta^2$, $\mathrm{Var}(\tilde{\mu}_1^{*}) = \kappa_0\sigma_\theta^2 + (1-\kappa_0)^2\sigma_\phi^2$, and $\mathrm{Cov}(\tilde{\mu}_1^{0*}, \tilde{\mu}_1^{*}) = \mathrm{Var}(\tilde{\mu}_1^{0*})$, so that $\mathrm{Corr}(\tilde{\mu}_1^{0*}, \tilde{\mu}_1^{*}) = \sqrt{\beta_{\mathrm{MZ}}} = \rho$. Appending the entry index $V_1 = \theta + \eta_1$ preserves joint normality, with $\mathrm{Var}(V_1) = \sigma_V^2 = \sigma_\theta^2 + \sigma_\eta^2$ and, since η_1 is independent of (s_0, ϕ),

$$\mathrm{Cov}(\tilde{\mu}_1^{0*}, V_1) = \mathrm{Cov}(\tilde{\mu}_1^{*}, V_1) = \mathrm{Cov}(\kappa_0(\theta + \varepsilon_0), \theta) = \kappa_0\sigma_\theta^2 = \mathrm{Var}(\tilde{\mu}_1^{0*}),$$

the martingale property of the Bayesian posterior mean, $\mathbb{E}[V_1 \mid \tilde{\mu}_1^{0*}] = \tilde{\mu}_1^{0*}$. With $v = \sigma_V/\sigma_{\tilde{\mu}_1^{*}}$, the correlation matrix of $(\tilde{\mu}_1^{0*}, \tilde{\mu}_1^{*}, V_1)$ is therefore

$$\Sigma: \qquad \mathrm{Corr}(\tilde{\mu}_1^{0*}, \tilde{\mu}_1^{*}) = \rho, \qquad \mathrm{Corr}(\tilde{\mu}_1^{0*}, V_1) = \tfrac{\rho}{v}, \qquad \mathrm{Corr}(\tilde{\mu}_1^{*}, V_1) = \tfrac{\rho^2}{v}, \tag{C-1}$$

which satisfies the Markov factorization $\mathrm{Corr}(\tilde{\mu}_1^{*}, V_1) = \mathrm{Corr}(\tilde{\mu}_1^{*}, \tilde{\mu}_1^{0*})\,\mathrm{Corr}(\tilde{\mu}_1^{0*}, V_1)$: conditional on the ϕ-free forecast, the latent forecast and the entry index are independent.

With $a = (c_A - \bar{\theta})/\sigma_{\tilde{\mu}_1^{*}}$ and $z_V = (c_E - \bar{\theta})/\sigma_V$, the standardized thresholds for $(\tilde{\mu}_1^{0*}, \tilde{\mu}_1^{*}, V_1)$ at (c_A, c_A, c_E) are $(a/\rho,\ a,\ z_V)$. Then,

$$p_{act}^{exc} = \Pr(\tilde{\mu}_1^{*} > c_A,\ V_1 > c_E) - \Pr(\tilde{\mu}_1^{0*} > c_A,\ \tilde{\mu}_1^{*} > c_A,\ V_1 > c_E) = \overline{\Phi}_2\Big(a, z_V; \tfrac{\rho^2}{v}\Big) - \overline{\Phi}_3\Big(\tfrac{a}{\rho}, a, z_V; \Sigma\Big),$$

$$p_{act}^{for} = \Pr(\tilde{\mu}_1^{0*} > c_A,\ V_1 > c_E) - \Pr(\tilde{\mu}_1^{0*} > c_A,\ \tilde{\mu}_1^{*} > c_A,\ V_1 > c_E) = \overline{\Phi}_2\Big(\tfrac{a}{\rho}, z_V; \tfrac{\rho}{v}\Big) - \overline{\Phi}_3\Big(\tfrac{a}{\rho}, a, z_V; \Sigma\Big),$$

where $\overline{\Phi}_2, \overline{\Phi}_3$ are the bivariate and trivariate standard normal survival functions. Subtracting and adding these expressions yields the formulas in Proposition 2; the common trivariate term cancels in the net measure, which therefore equals the difference between the actual-entry probabilities with and without the prior noise.

Appendix D Calibration of the entry-distortion measures

D.1 Data and calibration

The measures in Table 12 require the four standardized parameters (ρ, a, z_0, v), where $z_0 = (c_E - \bar{\theta})/\sigma_{\tilde{\mu}_1^*}$ is the standardized positive-expectation threshold and $z_V = z_0/v$. Taking the latent pool of potential entrants to be the U.S. resident population aged 24–64 (Pop), three of the four parameters are observed directly:

$$\rho^2 = \beta_{\mathrm{MZ}}, \qquad 1 - \Phi(a) = \frac{\mathrm{BA}}{\mathrm{Pop}}, \qquad 1 - \Phi(z_0) = \frac{\mathrm{WBA}}{\mathrm{Pop}}, \tag{D-1}$$

where β_{MZ} is the year-specific two-part model slope (Table 8), BA is total business applications, and WBA is applications with planned wages. The remaining parameter, v, is calibrated to the 4-quarter employer-transition rate of all applicants reported in Section 3:

$$\frac{\overline{\Phi}_2\big(a,\, z_0/v;\, \rho^2/v\big)}{1 - \Phi(a)} = 0.07. \tag{D-2}$$

The corresponding transition rate for WBA, $\overline{\Phi}_2\big(z_0,\, z_0/v;\, \rho^2/v\big)/\big(1 - \Phi(z_0)\big)$, is not targeted and serves as an overidentification check; the model-implied values, reported in the last column of Table D-1, range from 25% to 28% against the empirical 27%. Panel A of Table 12 feeds $\big(\beta_{\mathrm{MZ}},\, \mathrm{WBA/Pop}\big)$ into the closed forms of Proposition 1; Panel B applies Proposition 2. The "/app" columns use the observed application probability BA/Pop as the denominator.

Table D-1: Data moments and calibrated parameters

	Data				Parameters				
Year	Pop	BA	WBA	β_{MZ}	a	z_0	v	$\mathrm{Corr}(\tilde{\mu}_1^{0*}, V_1)$	WBA rate
2017	174,906,893	3,182,674	484,215	0.812	2.093	2.774	0.926	0.973	0.281
2018	175,384,903	3,486,829	503,783	0.806	2.056	2.762	0.932	0.964	0.279
2019	175,593,909	3,498,990	478,640	0.789	2.055	2.779	0.944	0.941	0.262
2020	175,013,298	4,356,498	524,651	0.768	1.962	2.748	0.964	0.910	0.254
2021	174,640,901	5,390,816	638,140	0.759	1.868	2.682	0.964	0.904	0.254

Notes. Pop is the U.S. resident population aged 24–64 (July 1 estimates; Census Bureau intercensal estimates for 2017–2019 and Vintage 2025 estimates for 2020–2021). BA is total business applications and WBA is applications with planned wages (Census Bureau, BFS). β_{MZ} is the year-specific two-part model slope from Table 8. The thresholds are $a = \Phi^{-1}(1 - \mathrm{BA/Pop})$ and $z_0 = \Phi^{-1}(1 - \mathrm{WBA/Pop})$; the implied application probabilities BA/Pop are 1.82%, 1.99%, 1.99%, 2.49%, 3.09%. v solves (D-2); $\mathrm{Corr}(\tilde{\mu}_1^{0*}, V_1) = \rho/v$. "WBA rate" is the model-implied 4-quarter employer-transition rate of WBA (not targeted; empirical value 0.27).

D.2 Application-margin distortions

The uninformative component of the priors also distorts the application margin, which is the channel through which the actual-entry distortions in Panel B of Table 12 arise. An agent is an *excess applicant* if she applies with the noisy forecast but would not absent ϕ, and a *forgone*

applicant in the reverse case. The probabilities of excess and forgone application are:

$$p_{app}^{exc} = \Pr\big(\tilde{\mu}_1^{0*} \le c_A,\ \tilde{\mu}_1^{*} > c_A\big)\,, \tag{D-3}$$

$$p_{app}^{for} = \Pr\big(\tilde{\mu}_1^{0*} > c_A,\ \tilde{\mu}_1^{*} \le c_A\big)\,. \tag{D-4}$$

Net excess application and *gross application distortion* rates are

$$p_{app}^{net} = p_{app}^{exc} - p_{app}^{for}, \qquad p_{app}^{dist} = p_{app}^{exc} + p_{app}^{for}. \tag{D-5}$$

These measures have the same structure as the expected-entry measures with c_A in place of c_E, so Proposition 1 applies with the expected-entry share replaced by the application share $r_A \equiv \Pr(\tilde{\mu}_1^{*} > c_A) = \text{BA/Pop}$:

$$p_{app}^{net} = \Phi\Big(\tfrac{a}{\rho}\Big) - \Phi(a), \qquad p_{app}^{dist} = \Phi\Big(\tfrac{a}{\rho}\Big) + \Phi(a) - 2\,\Phi_2\Big(\tfrac{a}{\rho},\, a;\, \rho\Big), \tag{D-6}$$

where $a = \Phi^{-1}(1 - r_A)$, and both measures are strictly decreasing in β_{MZ} since $r_A < \frac{1}{2}$. Table D-2

Table D-2: Application-margin distortion from heterogeneous priors

Year	β_{MZ}	p_{app}^{exc} (%)	p_{app}^{for} (%)	p_{app}^{net}/app (%)	p_{app}^{dist}/app (%)
2017	0.812	1.09	0.28	+44.4	75.7
2018	0.806	1.20	0.31	+44.7	76.0
2019	0.789	1.26	0.30	+48.1	78.4
2020	0.768	1.60	0.37	+49.4	79.4
2021	0.759	1.96	0.47	+48.2	78.9

Notes. $p_{app}^{exc}, p_{app}^{for}$ are pool-level probabilities in percent of the population aged 24–64. The "/app" columns divide the net ($p_{app}^{exc} - p_{app}^{for}$) and gross ($p_{app}^{exc} + p_{app}^{for}$) distortions by the observed application probability BA/Pop, giving distortion in percent per applicant. Parameters are those of Table D-1.

shows that net excess applications are positive throughout, at 44–49% of observed applications per year, with gross distortion of 76–79% per applicant: belief noise substantially inflates the volume of filings relative to the ϕ-free counterfactual. Combined with Panel B of Table 12, these figures yield the conversion asymmetry underlying the negative net actual-entry distortion: applicants induced by noise attain employer status at rates below one percent ($p_{act}^{exc}/p_{app}^{exc}$), whereas those deterred would have converted at rates approaching the applicant-pool average ($p_{act}^{for}/p_{app}^{for} \approx$ 3–8%, against a 7% baseline).